



P4P Scribbles

See document tabs

<https://www.overleaf.com/project/6a9a0b15a01c0ae915d3d2f3>

Check ipad notes

Hemolysis

- Hemolysis index
- Normalized index of Hemolysis
 - ASTM F1841 Standard, for testing TAH hemolysis (costs money :() **BUT IS ABOUT REAL BLOOD SO GAF**
 - [Hemolysis testing in circulatory support pumps: Standards, mock loops, and pulsatility](#)
- Heuser-Opitz (HO) power law model for shear for % hemoglobin damaged

$$HI(\%) = C \cdot \tau^\alpha \cdot t^\beta$$

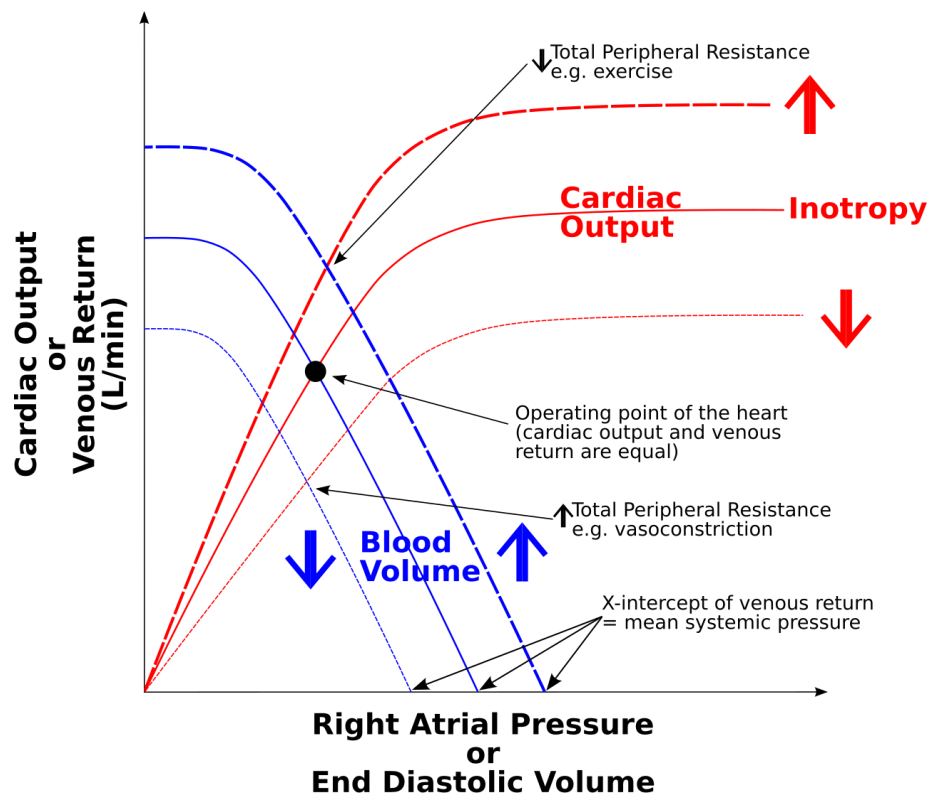
- non-newtonian carreau yasuda equation blood (governing equation for the fact blood gets less viscous the more it's sheared) ??? Tarusha
- <https://www.sciencedirect.com/science/article/pii/S2405844025010151>
- <https://pmc.ncbi.nlm.nih.gov/articles/PMC8106046/> - **INSANELY GOOD SIM ARTICLE**, they find a three-dimensional shear stress tensor from their sims for blood. They **DONT** use the non-newtonian carreau yasuda equation.
 - 30% Hematocrit
 - NIH larger than 0.1 g/100 L indicates hemolysis in the blood pump

NIH = HI% * constant * (1 - Hct) * Hb is how we convert from HO power law to NIH

[Fluid-structure interaction modelling of a positive-displacement Total Artificial Heart](#)

- For MECHANICS of heart
- <https://pubmed.ncbi.nlm.nih.gov/42010686/> Syncardia vs Realheart hemolysis
 - 37.15 ± 12.42 mg/100 L is the Syncardia blood damage
- <https://www.syncardia.com/download/consultation-guide-for-the-syncardia-t-tah.pdf>
 - Stroke volume of 50cc Syncardia is 50mL, obviously, but is underfilled to between **30-40mL** ([Link](#))([Link](#))

Starling's Law is the pressure relationship with fluid volume in a heart



READ MORE INTO https://en.wikipedia.org/wiki/Cardiac_function_curve

https://media.licdn.com/dms/image/v2/D4E22AQEIXZfXwGtKoQ/feedshare-shrink_1280/B4EZtNtRjBIsAw-/0/1766535285174?e=1781136000&v=beta&t=7ZTjS-ghT9BlddjwkAR D4W3Xf2_fGoEZSC4ckscVXI0

- Tharusha's hemolysis poster

The NIH is calculated using the following formula:

$$\text{NIH} = \Delta Hb \times V \times \frac{(100 - Hct)}{100} \times \frac{100}{Q \times T}$$

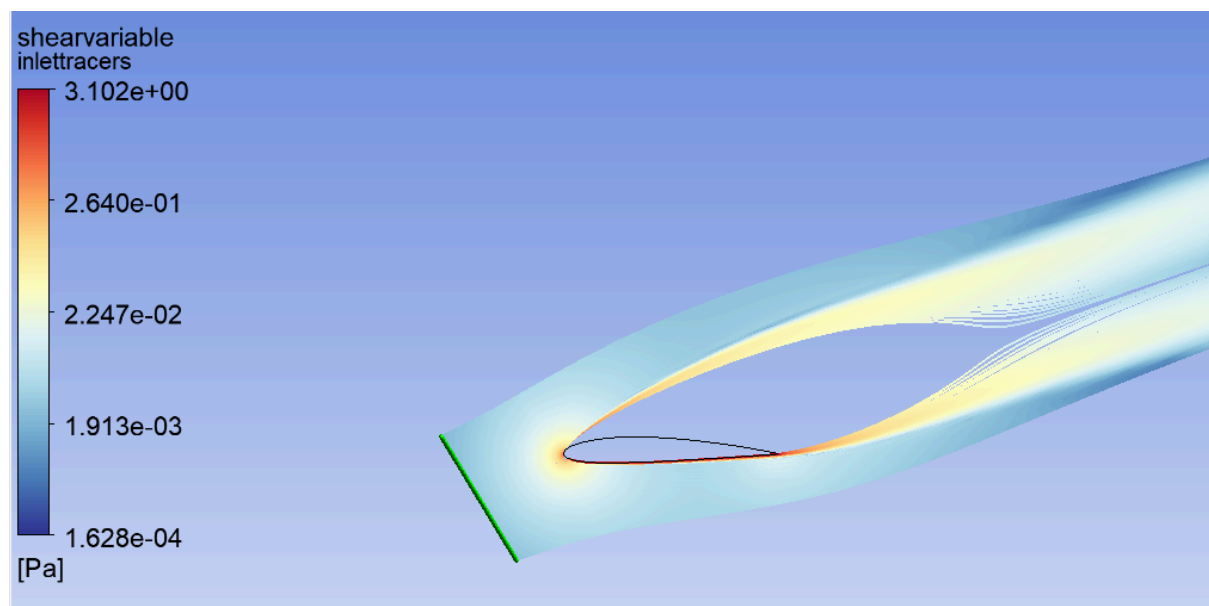
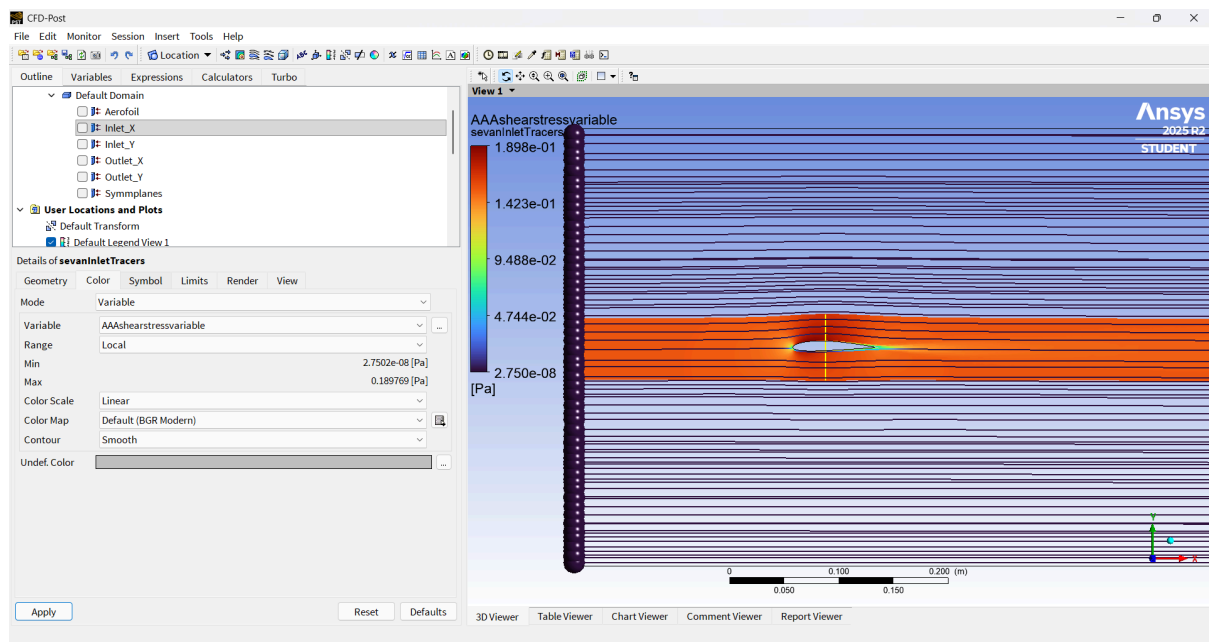
Where:

- ΔHb : Change in plasma-free hemoglobin concentration (g/L)
- V : Volume of the circulation loop (L)
- Hct : Hematocrit percentage
- Q : Pump flow rate (L/min)
- T : Circulation time (min)

https://ideaexchange.uakron.edu/cgi/viewcontent.cgi?article=2257&context=honors_research_projects

Programmatical solution for hemolysis 10/08/2026 through to 18/08/2026

See [mid-year technical report](#)



Streamlines 😊!!!

The workflow here is

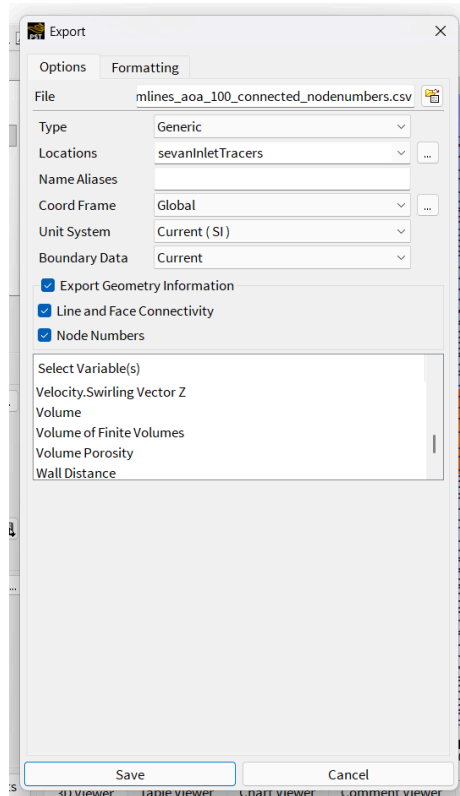
- add **Expression** defined as **Shear Strain Rate * Dynamic Viscosity**
- defining a new variable with that expression
- define a line in front of the area of interest for the streamline to originate
- add inlet tracer 3D streamline set originating at the inlet line (perpendicular to flow)
 - Set that newly created variable as the **Color** information applied via **Mode Variable**

Export dialogue

Location is the streamline

- AAAshearstress (custom variable)
- X,Y,Z
- Velocity

with **NODE NUMBERS** and **CONNECTIVITY** selected



CLANKER CODE ADVICE::

Parse [Data] into a DataFrame indexed by node number (X, Y, Z, shear stress, velocity)

Parse [Lines] and walk it, splitting into separate streamlines wherever the chain breaks (start-node $\neq$ previous end-node)

For each streamline: compute segment length between consecutive nodes, divide by local velocity to get Δt , accumulate cumulative time

Apply your discretized equation (2) per streamline, get HI% per line

Flow-weight and average across all streamlines $\rightarrow$ device HI% $\rightarrow$ convert to NIH via equation (3)

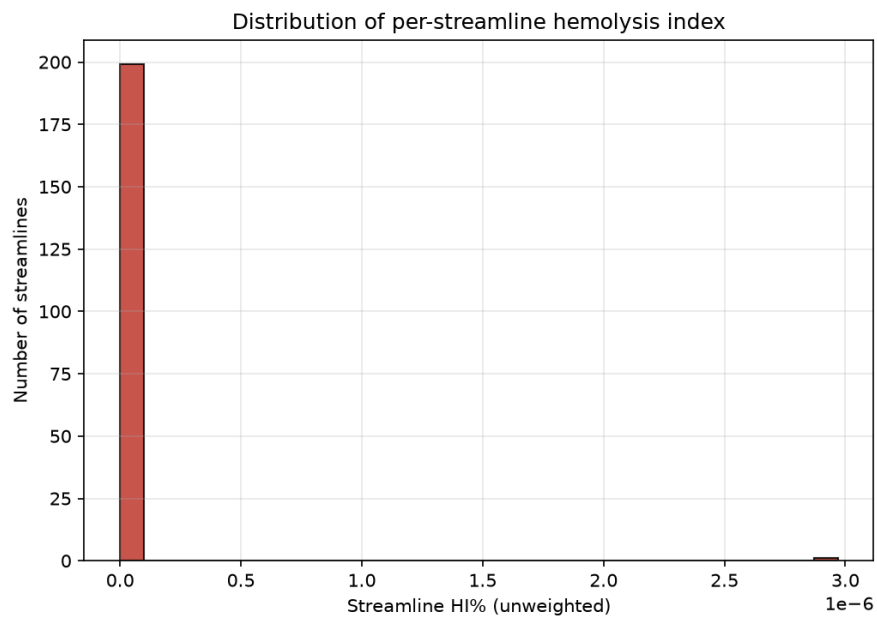
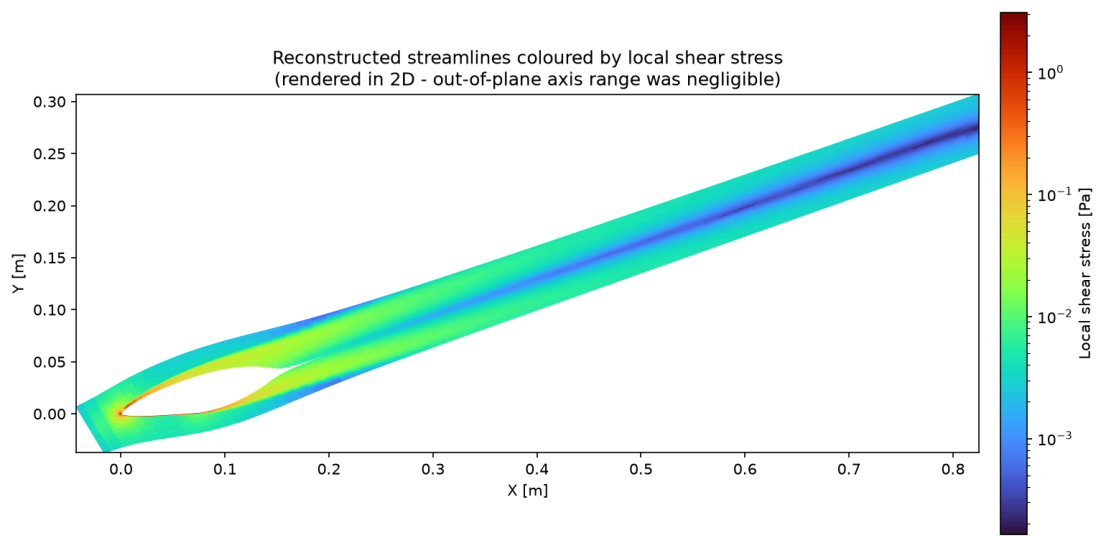
Outputs of this code:

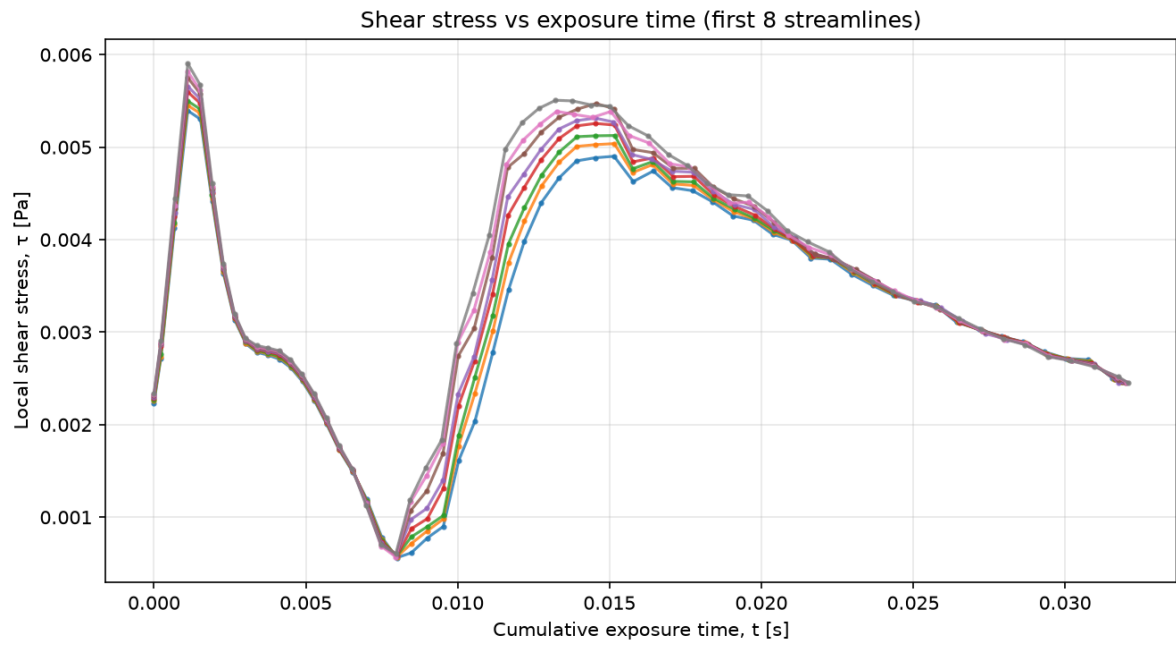
```
=====
Streamlines processed:          200
Device HI% (flow-weighted):    1.496214e-08 %
Device NIH:                    0.0016 mg/100L
SynCardia 50cc literature:     37.15 +/- 12.42 mg/100L
ASTM acceptable limit:        100.0 mg/100L
=====

=== Streamline Quality Check ===
Total streamlines: 200
Average points per streamline: 61.165
Min/Max velocity: 2.713 / 37.860 m/s
Min/Max shear: 0.0 / 3.1 Pa
Saved: streamlines_3d.png, shear_vs_time.png,
hi_histogram.png, streamline_summary.csv
```

The extremely low haemolysis and shear values make sense considering that this is an aerofoil and not a heart

Below are the plot outputs:





Title Page

Acknowledgements

Abstract/Executive summary

Table of contents —

Introduction

Literature review

----- **Narrative trail to what we ended up actually doing ?**

Methods

Brief Summary of Methods

Design Overview

SynCardia 50cc Introduction & Assumptions

35cc SynCardia Parameters

Downsizing Explained

(Parameters like gradation, leaflet valves, etc)

Hemolysis Evaluation

Physical Rationale / Purpose

Programmatical Solution

Simulation Design / Parameters

Simulation Purpose *(Doing what for why)*

Mesh Justification

2D Steady-State

3D Steady-State

Transient *(If applicable)*

Presentation of results

Preliminary vs. Downsized 3D Steady-State Results

Individual Parameters Tested in Isolation Results

Hemolysis Results

Transient *(If applicable)*

Discussion of findings

Viability as a TAH *(Physiology vs. Flow Results)*

Physical Trends / Behaviours

Hemolysis Trends / Behaviours on Downsizing

Discussion of improvements

Conclusion

Discussion of Future Work

List of references

Appendices

Talk about IPMCs in background >????

Population stuff

Carreau-yasuda ignorance

Loop

- Support **5 L/min flow rate**, *Cardiac Output at rest, not normalized with Cardiac Index*
- Blood enters heart non-pulsatile/continuous -> reservoir before heart
- If simulating a pulmonary loop, **2-5% less volume output from the right ventricle/to that loop**

Heart

- **Mean Arterial Pressure target** is **120-70 mmHg**, that's pressure body-wide (**16kPa - 9.3 kPa**) but **65-85mmHg in a 50cc Syncardia running at 180-200mmHg for systole in left ventricle.**
- **Stroke volume** between **30-40mL** if basing on 50cc Syncardia (max 50mL)
 - A starting 'guess' for the specific stroke volume, as well as ventricle balance that our TAH must operate at can be informed by the **Frank-Starling law**
- **Hemolysis reference:** **37.15 ± 12.42 mg/100 L** based on Syncardia 50cc
 - NIH larger than 0.1 g/100 L indicates hemolysis in the blood pump
 - **$NIH = HI\% * constant * (1 - Hct) * Hb$** is an equation I vibed out based on HI% power law being % damaged hemoglobin, Hct being the percentage of blood that is red blood cells, and Hb is hemoglobin concentration (150g/L)

Welcome to lit review ongoing, which is where I'm collating studies relating to simulation parameters in CFD, downsizing methodologies, and basically anything that relates to our newer research question

Evaluation-of-eulerian-and-lagrangian-models-for-hemolysis

<https://www.ovid.com/jnls/asaiojournal/pdf/10.1097/mat.0b013e318254833b~evaluation-of-eulerian-and-lagrangian-models-for-hemolysis> Directly Important ▾ Read ▾ 7 Oct 2026

- See [P4P SCRIBBLES](#) Ongoing tab, it's fantastic for hemolysis modeling

[Numerical Model of Full-Cardiac Cycle Hemodynamics in a Total Artificial Heart and the Effect of Its Size on Platelet Activation](#) Directly Important ▾ Reading ▾ 4 Sept 2026

- Length scaling factors on 50cc to achieve 35cc
- Uses stress accumulation model, NOT power law
 - $SA = \sum \tau \cdot dt$ equiv.
 - ★ **Implement the simpler form in my code** ✓
- Medtronic Hall Valve no longer in use but previously was used in SynCardia, now Syn-Hall

[Requiem for a good mechanical heart valve: Farewell to the Medtronic Hall valve](#)

Maybe Helpful! ▾ Reading ▾ 4 Sept 2026

- Has Medtronic Hall valve figure



FIGURE 1. Medtronic Hall valve (Easy-Fit model) (Medtronic Inc, Minneapolis, Minn).

The Medtronic-Hall valve: a design in 1977 to improve the results of valve replacement

Maybe Helpful! ▾ Not Read ▾ 4 Sept 2026

- Specifies the valve quantitatively

https://www.researchgate.net/publication/250920158_Total_artificial_heart_in_the_pediatric_patient_with_biventricular_heart_failure

Maybe Helpful! ▾ Not Read ▾ 6 Oct 2026

- Has flow graph of SynCardia

<https://pmc.ncbi.nlm.nih.gov/articles/PMC4854792/pdf/nihms-769456.pdf>

Repetitive Hypershear Activates and Sensitizes Platelets in a Dose-Dependent Manner

Directly Important ▾ Not Read ▾ 6 Oct 2026

- No statistically significant platelet activation for single pass, but was important on multiple exposures
 - ***LIMITATION OF OUR RESEARCH***

https://www.researchgate.net/publication/270513851_Physiological_Characterization_of_the_SynCardia_Total_Artificial_Heart_in_a_Mock_Circulation_System

Physiological Characterization of the SynCardia Total Artificial Heart in a Mock Circulation System

Directly Important ▾ Not Read ▾ 6 Oct 2026

- Mentions afterload and number justifications
- ***TAH is insensitive to variations in afterload YAYAYAYAY IMPORTANT !!! JUSTIFIES OUR AFTERLOAD !!!!! AWINFDOAWSFONIFONAIOFNI***

Repetitive Hypershear Activates and Sensitizes Platelets in a Dose-Dependent Manner

<https://pmc.ncbi.nlm.nih.gov/articles/PMC4854792/pdf/nihms-769456.pdf>

Yes—**this paper adds some very relevant discussion, especially around what your platelet SA results can actually establish.** It contributes more to your report's interpretation and limitations than to your haemolysis equations.

I read it against `haem5.py`, your empirical-threshold script, and `HAEM_DIAGNOSTICS.md`. I haven't seen your actual report, so "new" here means beyond those files. The paper is Sheriff et al. (2016), *Repetitive Hypershear Activates and Sensitizes Platelets in a Dose-Dependent Manner*.

1. Equal stress accumulation does not necessarily mean equal platelet activation.

Your SA calculation reduces a trajectory to $SA = \int \tau dt$. The paper discusses substantially different activation responses at similar SA when stress magnitude, waveform and repetition differ. Its comparisons with earlier experiments suggest that intermittent high stresses can produce more activation than constant exposure with a similar total dose. **PDF pp. 7–8.**

For your report, this supports treating SA as a useful exposure and design-comparison metric, rather than a complete biological activation model. Two geometries could have similar SA distributions but different peak stresses and exposure patterns.

For your code, it gives a reason to examine **peak stress and stress histories alongside SA**—much of which your diagnostics already supports.

2. Platelets can retain a response to previous shear exposure.

The paper investigates *sensitization*: platelets exposed to high shear may subsequently continue activating in low shear. This is especially relevant when a trajectory ends at your device outlet; biological consequences may continue downstream. **PDF pp. 6, 8–9.**

However, **the single-passage sensitization trend in this experiment was not statistically significant.** Present it as suggestive evidence and a limitation of your current model, rather than a proven outcome for your TAH.

Your code accumulates exposure but does not model platelet activation state or sensitization. That distinction deserves a paragraph in the report.

3. A single device passage cannot establish the effect of repeated circulation.

Repeated exposures produced significant activation in the variable-shear experiment. This adds biological motivation to the repeated-exposure limitation already acknowledged in your empirical screening notes. **PDF pp. 6–9, Table 1.**

Two different effects matter:

- Recirculation inside the device can prolong a particular passage.
- Systemic circulation can bring previously exposed platelets back through the device.

Your streamline analysis may capture the first within its tracking limits; it does not establish the second. Multiplying single-pass SA by a passage count would give an exposure estimate, but would not reproduce sensitization or recovery between passages.

4. This paper makes your fixed platelet “threshold” wording too strong.

The repeated-exposure experiment first found significant PAS elevation at **1611 dyne·s/cm²**, under its particular protocol. That is **not a replacement for your 35 dyne·s/cm² Hellums reference**. The endpoints, blood preparations and exposure histories differ. **PDF pp. 5–8.**

More importantly, your code currently describes the Hellums line as a “clinical activation threshold” and implies that the fraction exceeding it directly estimates risk. This paper supports more cautious wording:

“Fraction of sampled trajectories exceeding the selected SA reference criterion.”

That fraction is neither a measured fraction of activated platelets nor a clinical thrombosis probability. I would prioritise correcting that interpretation before adding another model.

5. Platelet activation, platelet lysis and haemolysis are different endpoints.

The study measures platelet procoagulant activity through a modified prothrombinase assay, plus morphology. It does **not** validate your RBC haemolysis HI/NIH predictions or your platelet-lysis boundary. **PDF pp. 5, 9.**

Its experiments also used isolated platelets at low concentration, with short intervals between repeated exposures. The authors explicitly caution that this differs from physiological blood and circulation. Those limitations prevent directly transferring its experimental threshold to your CFD results.

My recommendation: use this paper prominently to discuss **the limits of SA, repeated exposure, and platelet memory**. Keep your existing haemolysis calculations; the most immediate improvement is to qualify the platelet threshold language. The key report argument is that your analysis compares mechanical exposure, while biological thrombogenicity additionally depends on exposure history and the response being measured.

All page references above refer to your local PDF (papers/nihms-769456.pdf).

 P4P SCRIBBLES 25th May

25th May Meeting Notes

- Talk about hemolysis and heart specs  P4P SCRIBBLES
 - I Read heart hackathon info again
 - We really want Tharusha in the Teams chat
 - 35cc article - have not read it yet
 - We were supposed to appear with a sketch this meeting
 - I have target parameters, some based 50cc Syncardia, but no sketch
 - I want to read all the articles I found this week's sections on testing loops
 - Tharusha has informed me we can probably just use his heart loop as a base for designing ours or just using his
-
- Goal
 - We really need to thoroughly read 35cc paper for research purpose
 - Find NIH acceptable limits of hemolysis in g/100L
 - EMAIL raj our progress and thoughts during this week so we can move on to new things DURING the week

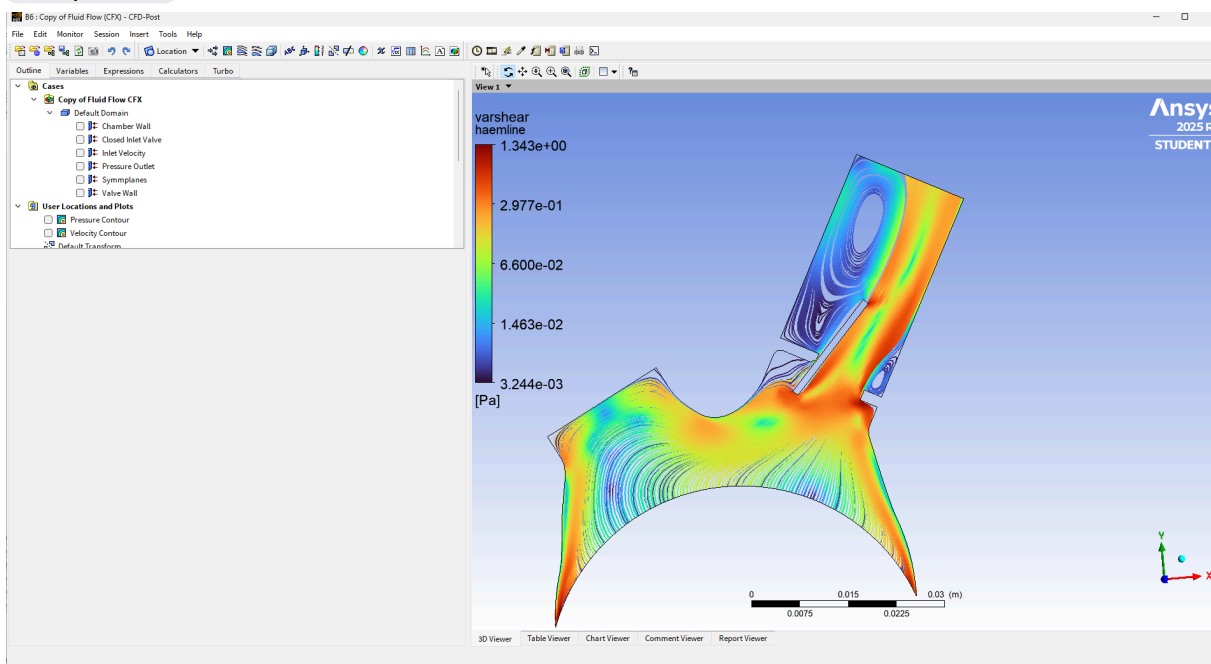
4 Sept 2026

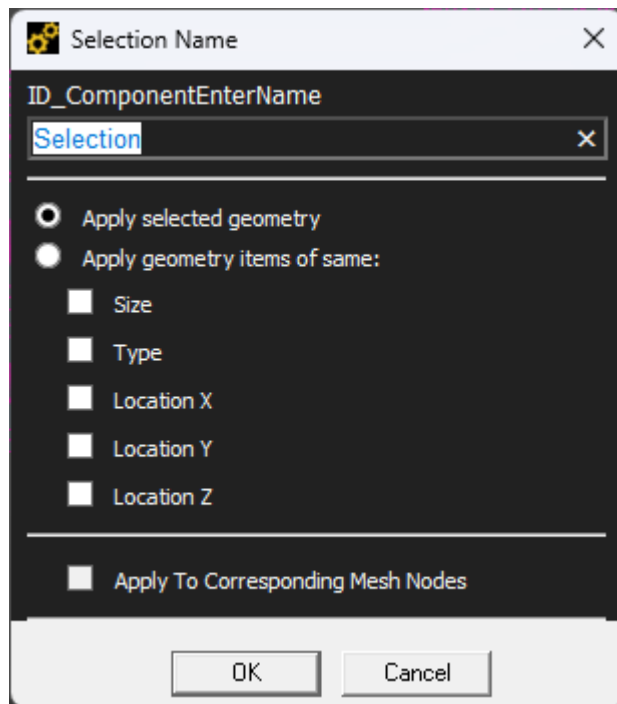
Created [P4P SCRIBBLES](#) *Lit Review Ongoing* tab and looked back into new literature on Scopus with the bangs TITLE-ABS-KEY("computational fluid dynamics" AND ("syncardia"))

Found patent WO2009084027A2 which may be for the Syn Hall valve, the [google patent](#) was broken so an [archive.is snapshot](#) was made to reveal it ([patent download](#))

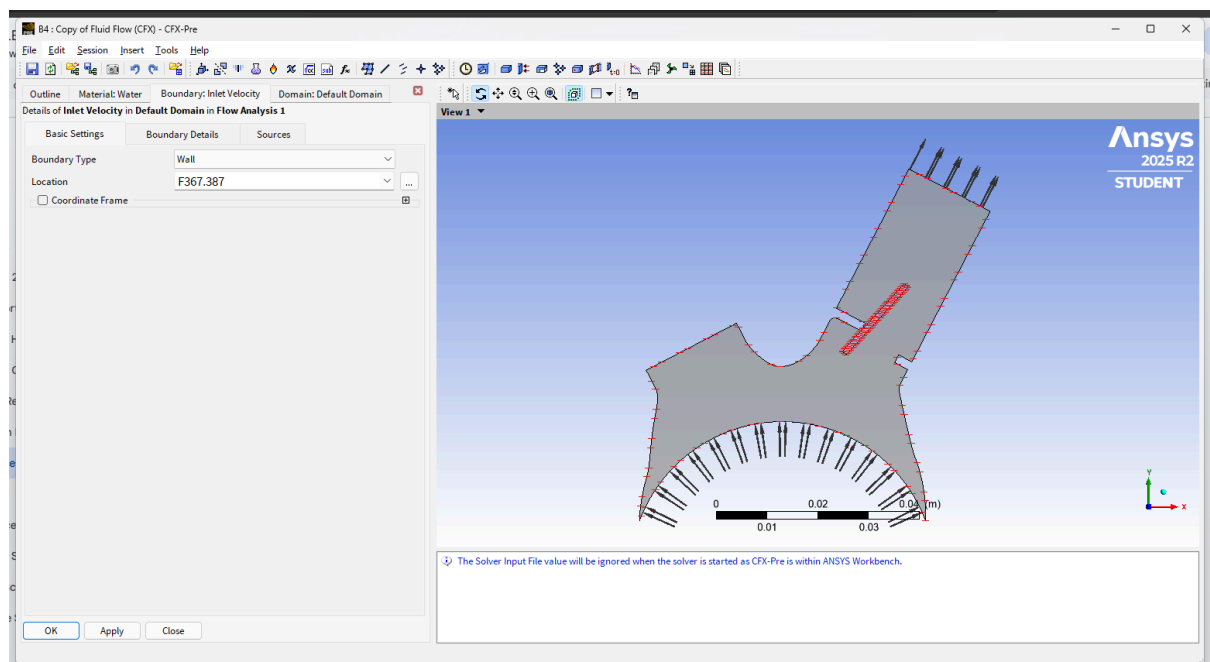
Found specification of Medtronic Hall Valve: The Medtronic-Hall valve: a design in 1977 to improve the results of valve replacement

9 Sept 2026





In mesh



Issues: Lack of named selections in the Mesh component means the boundary conditions aren't inferred by ansys, moreover is the diaphragm supposed to be an inlet ? the simulation breaks when changed to a wall

It's also a pain to not have them be named :(

1.3 Pa is too little for the max scalar shear stress measured at systole

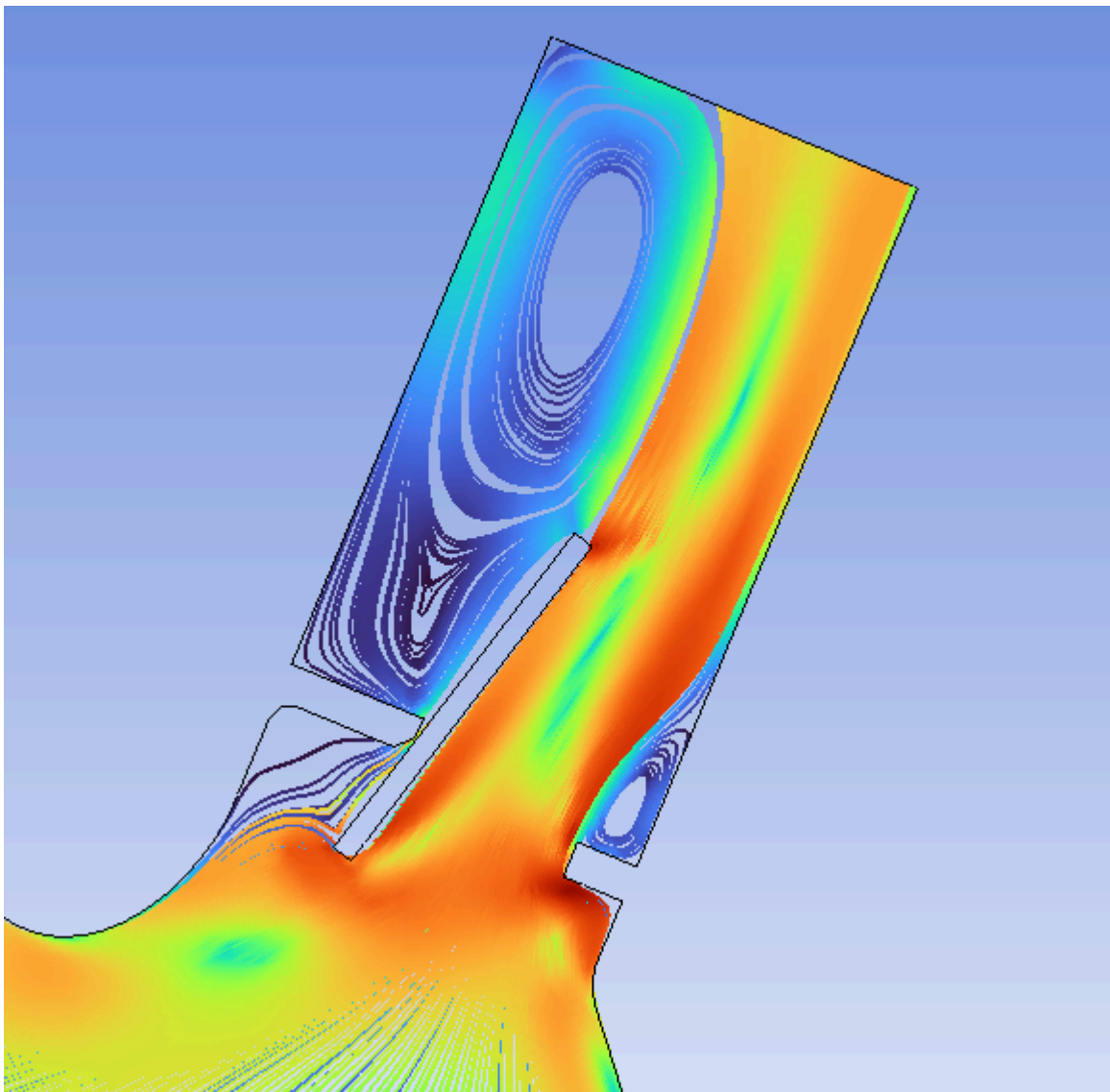
I am not including wall shear because I think scalar shear is more important and easier

Blood is defined as a custom material - I need to check its bulk modulus and other definitions

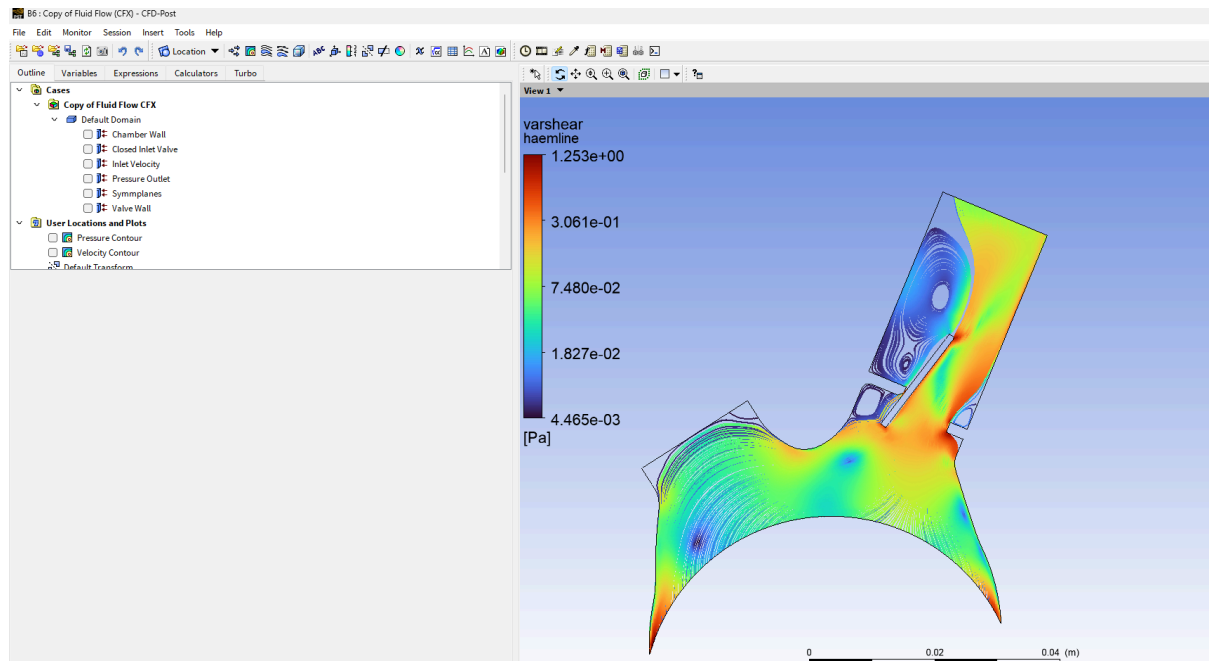
Why does the velocity inlet (diaphragm) feed at 0.042m/s (4.2 cm/s) ?

```
***** Notice *****  
A wall has been placed at portion(s) of an OUTLET  
boundary condition (at 37.5% of the faces, 27.8% of the area)  
to prevent fluid from flowing into the domain.  
The boundary condition name is: Pressure Outlet.  
The fluid name is: Blood.  
If this situation persists, consider switching  
to an Opening type boundary condition instead.
```

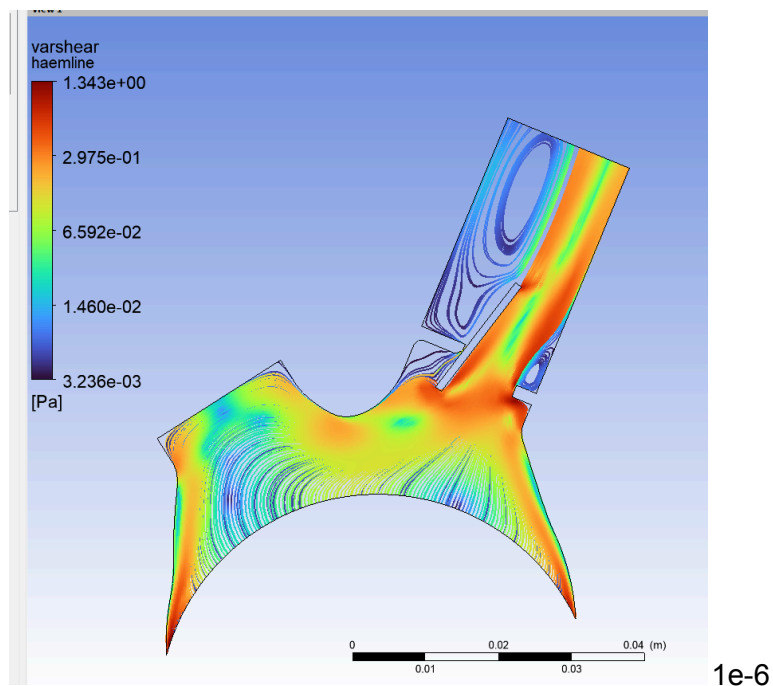
The outlet is a wall /// ?????? thats causing recirculation!!! and thats not !!!! accurate !!!!



like thats recirculation



Changed the outlet to an **opening**, and ran simulation at new convergence criteria of $1e-4$ because i dont have time to wait



$1e-6$

Blood properties:

Molar mass can remain the default value because I don't think it's used to calculate anything for the simulation?

Density being changed from 1050kg/m^3 to 1060kg/m^3
<https://www.sciencedirect.com/topics/engineering/human-blood>

Dynamic Viscosity from 0.004 to 0.0035 Pas (3.5 centipoise, i cannot for the life of me find the studies that mentioned this)

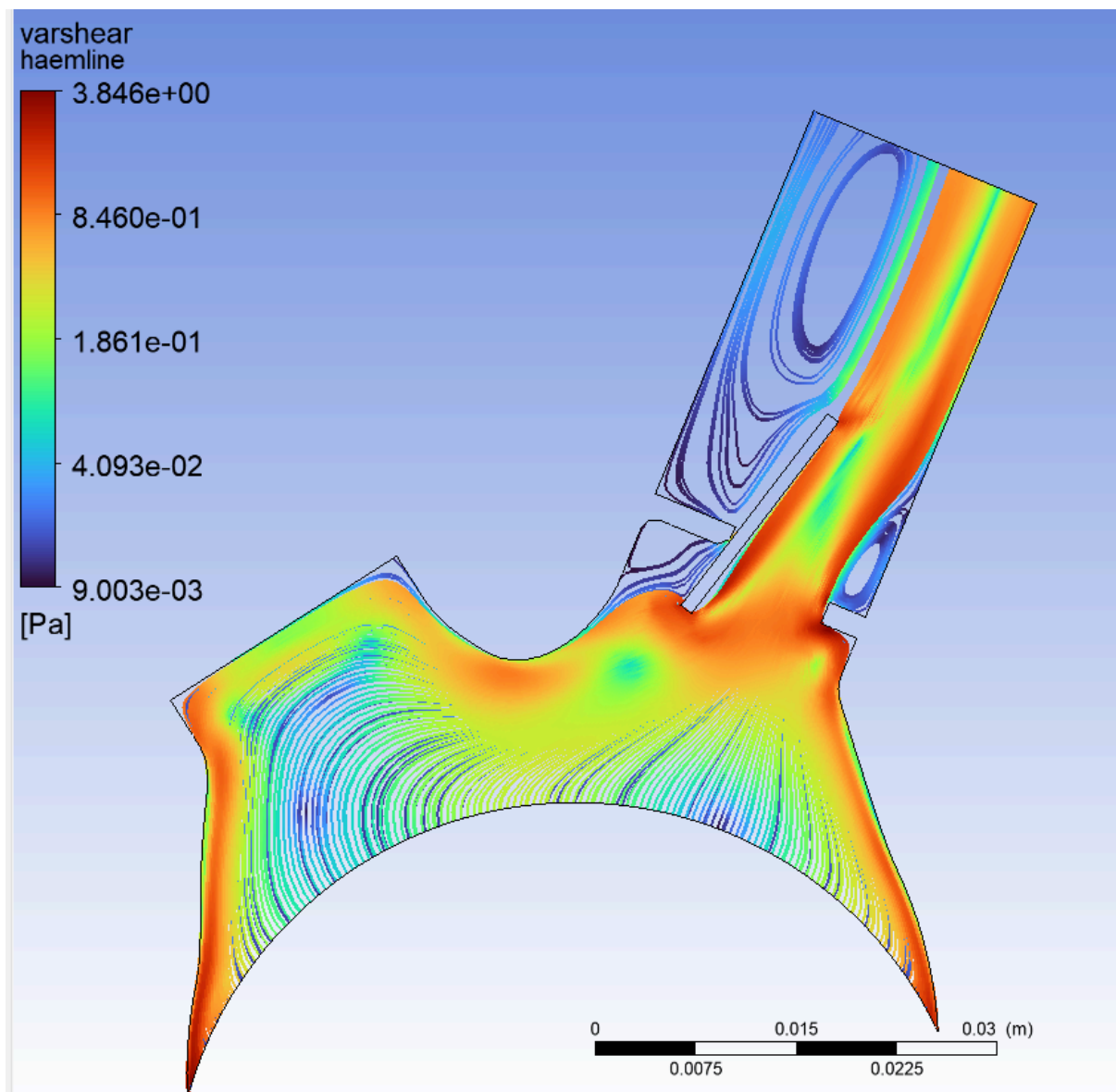
It can be incompressible because the blood is not moving fast enough for it to matter - $2.7 \cdot 10^9$ Pa is defined as the bulk modulus by [this paper](#) indicating it's mildly compressible but I don't think it matters, they cite a paper for [another number that models blood as incompressible anyway](#) (+ laminar + newtonian)

Velocity of the inlet:

4.5cm/s currently, but to cover the diaphragm distance (0.02 metres) at 120bpm is 12.6cm/s for mid systole assuming a sine wave like in the [study](#)

Backpressure of the outlet/opening:

100mmHg, no idea what it should be. Claude says 120mmHg but i dont know



New simulation at 12.6cm/s diaphragm motion, peak velocity is now 1.5m/s, averaging around 0.8 m/s at the outlet which is ... sort of ... normal ??? WOW !!!

We should probably double check the mesh resolution near the walls (y^+ according to google) since the shear is still low ? but it is also 2D so idk

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3D simulation investigation

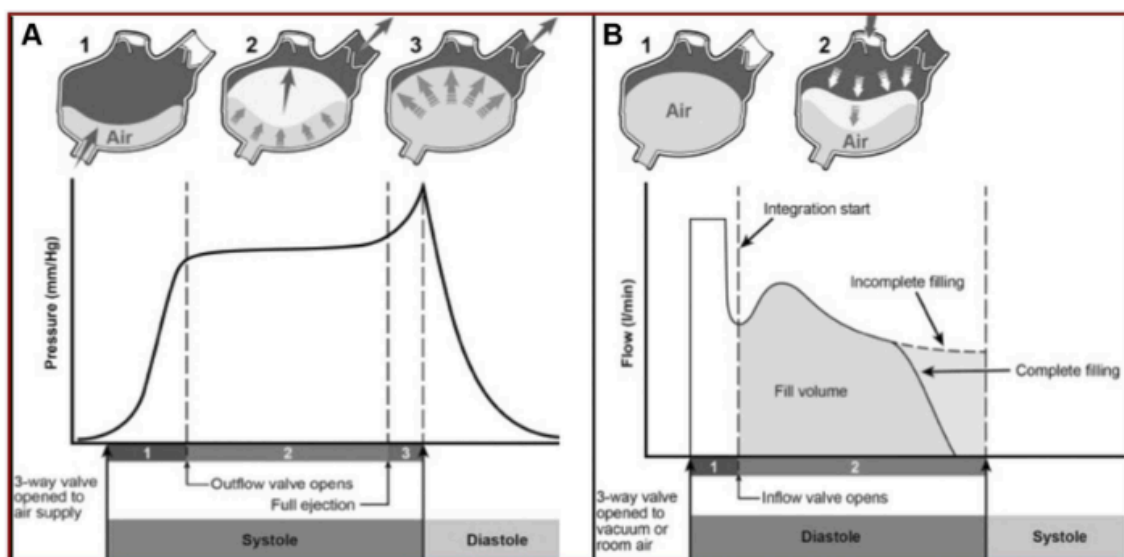
Find out why my hemolysis is underreported

Figure - available from: [Perfusion](#) ✓

This content is subject to copyright.

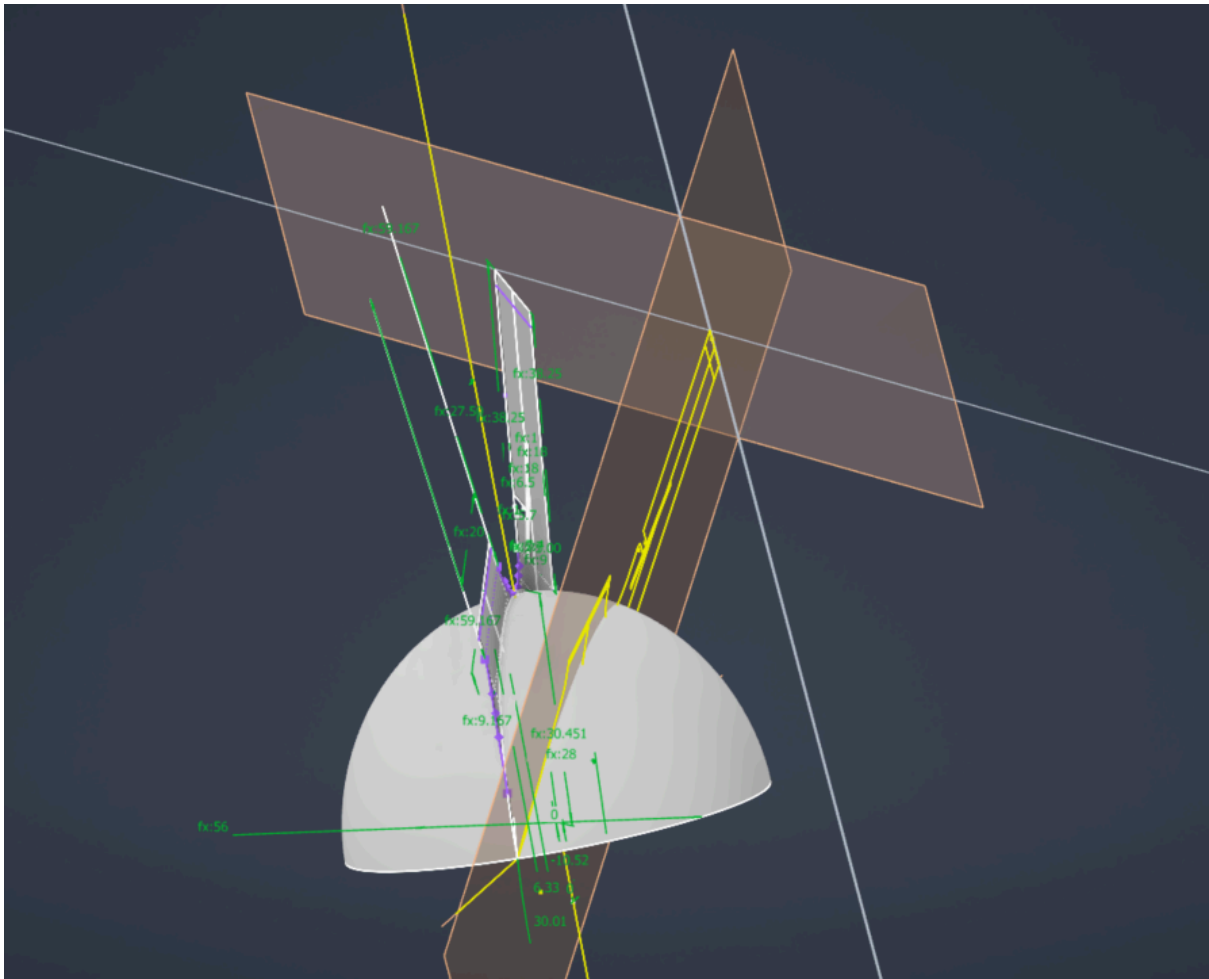
Download

View publication



The Cardiac Cycle of the SynCardia TAH-L. The cardiac cycle of the SynCardia Total Artificial Heart operates by shuttling pressurized air in and out of the pneumatic chambers of the left and right ventricles in synchronization through the attached percutaneous drivelines connected to the external driver and console. Figure 2A displays the systolic phase at the beginning when air is applied from the external driver (1) to the pneumatic chambers of the ventricles. When the sub-diaphragm ventricular pressures exceed the pulmonary and aortic pressures, the 25 mm outlet valves will open (2) and the diaphragms will displace upward until they reach maximum distension and all ventricular blood is ejected (3). The build-up of pressure is measured and the active tracing is displayed on the driver monitor. The accumulated pressurized air inside the ventricles is then released back along the drivelines to the console and exhausted into the atmosphere at the end of the systolic cycle and the beginning of the diastolic filling cycle. Transition from the systolic to diastolic phase is shown in Figure 2B; air is exhausted from the pneumatic chambers (1) of the ventricles back to the console to allow room for blood to fill the ventricles. Ventricular filling begins when the sub-diaphragm ventricular pressures decrease below the right and left atrial pressures and the 27 mm inlet valves open to allow blood to displace the diaphragms downward (2). The resulting displaced air along the drivelines from ventricular filling is measured by the console and estimates ventricular stroke volume. The active tracing is displayed on the driver monitor. A small vacuum which draws up to 15 mmHg during diastole can be applied from the console to help in diaphragm displacement and facilitate ventricular filling. Images Courtesy: syncardia.com

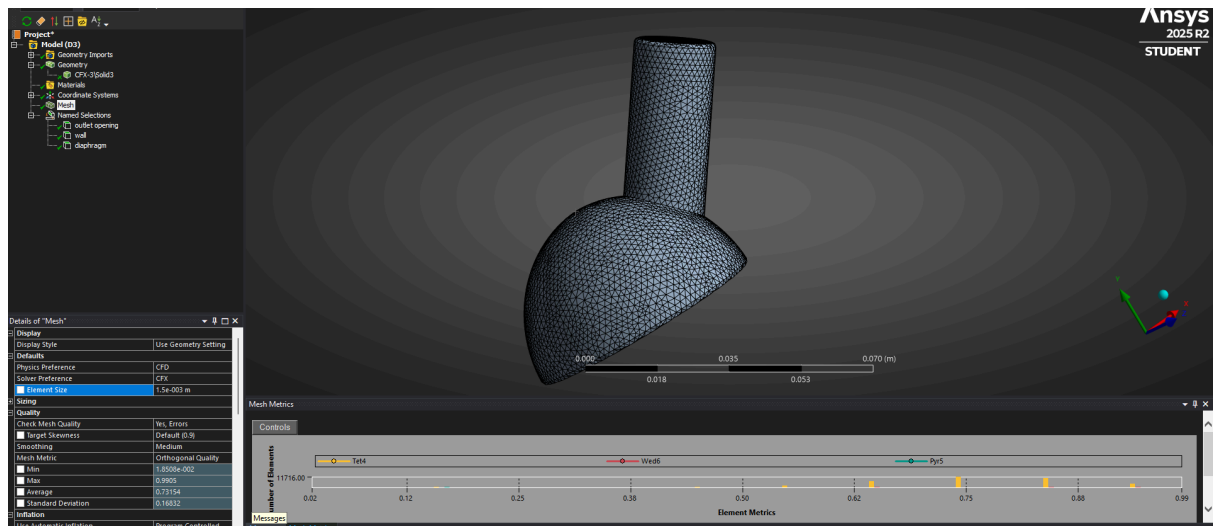
Oh my god ^^ [this paper](#)



Angling the inlet/outlets based on inference from the [university of dakron paper](#) below

“Mesh independence: since you're already comparing against a literature hemolysis value, a basic mesh sensitivity check (2–3 refinement levels, compare hemolysis index convergence, not just velocity) is worth having in your report — reviewers will ask.”

- From Claudius



Orthog error goes to $1.85e-2$ minimum, but on average is 0.74

Todo : expand the distance between the inlets and outlets



It didnt simulate :(

17 Sept 2026

Lots of git
100k hemolysis sim
Add loadbar
Meeting
Read papers

Evaluation-of-eulerian-and-lagrangian-models-for-hemolysis

<https://www.ovid.com/jnls/asaiojournal/pdf/10.1097/mat.0b013e318254833b~evaluation-of-eulerian-and-lagrangian-models-for-hemolysis>

- Does my haem model account for long-exposure time low-shear haemolysis ? (eq 3) **yes I'm using HI4, notably weak for undershooting expected haemolysis when the shear stressors are later on in time from $t=0$, the cumulative sum of damage favors earlier stress because it is the beta-1 denominator**
 - Eq 7 outlines a mechanical dose model that accounts for short and long term per-particle shear
- They use multiple reference frame approx for quasi steady state
- Hemolysis HI%total is averaged among all outflowing paths
- Number of pathlines didnt really change hemolysis index !!!!!!!!!!!
- Time steps didnt really change HI% !
- Which accumulator model am i using ?? and which does the study like the most (HI4 is mine)
-

Grill claude on why my pdf doesnt look right

Understand stress accumulator better

Try recreate plotting methods seen in the studies i.e. numerical methods

Report : one comparison plot possible could be power law comparison (three different models exist of the power law coefficients) and see which is the most similar to literature !!

18 Sept 2026

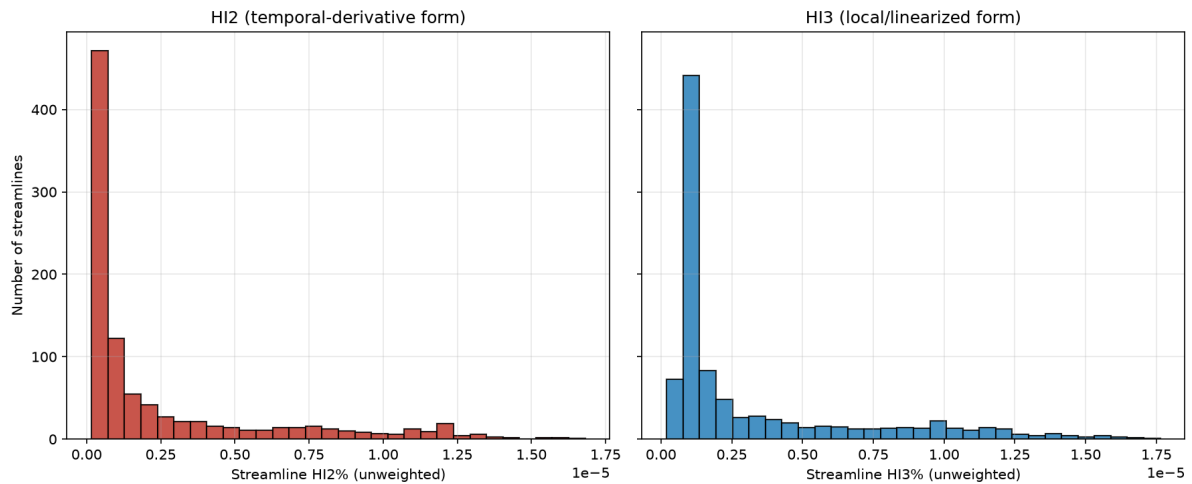
Evaluation-of-eulerian-and-lagrangian-models-for-hemolysis

<https://claude.ai/chat/0816cab3-36e5-40ed-9264-8d33c832a0bf> (equation explainer for HI4)

- HI4
- The paper notably says that they OMIT reynolds shear stresses (turbulence) as they elevate shear fields, we use SST k-epsilon or something and that does have a major turbulence component - its ok that i dont touch it :)
- *"since HI scales as r^α with $\alpha \approx 2$, any smoothing of the peak wall shear stress from a coarse near-wall mesh gets squared into a much larger HI error."* from admittedly claude (forgive me, I am tired today), acknowledges that in the absence of reynolds shear the mesh fineness near the walls becomes really really important for HI% estimation
 - y+ checkover in highest fineness regions is super duper important for the HI model to work properly !!!

Debugging:

My code underestimates hemolysis by a lot. I've since changed the model to HI3 from the study above, because it doesn't diminish the influence of shear at $t > 0$



As we can see, these are rather similar but the HI3 model overall produces greater HI%

HI2 (temporal-derivative form, Taskin et al. Eq. 4):

Device HI2% (flow-weighted): $2.652783 \times 10^{-6} \%$

Device NIH (from HI2): 0.2785 mg/100L

HI3 (local/linearized form, Taskin et al. Eq. 5):

Device HI3% (flow-weighted): $3.305893 \times 10^{-6} \%$

Device NIH (from HI3): 0.3471 mg/100L

Still both not great !!! Too low overall !!!

Also I added Giersepen constants, and Heuser-Opitz constants as separate things I can switch between, I mislabeled them in previous versions.

aomsdsamdpmpad

```
=====
==
Streamline coverage diagnostic
=====
==
Streamlines analysed: 958
Segments floored (all lines): 0.00 % (median per-line:
0.00 %)
Streamlines with >10% floored: 0.00 %
Streamlines with >50% floored: 0.00 %
-----
--
Total transit time t_total [s]:
min / median / max: 2.2293e-04 / 9.6095e-02 /
2.3217e-01
std / mean: 4.9896e-02 / 1.2538e-01
coefficient of variation: 0.398
```

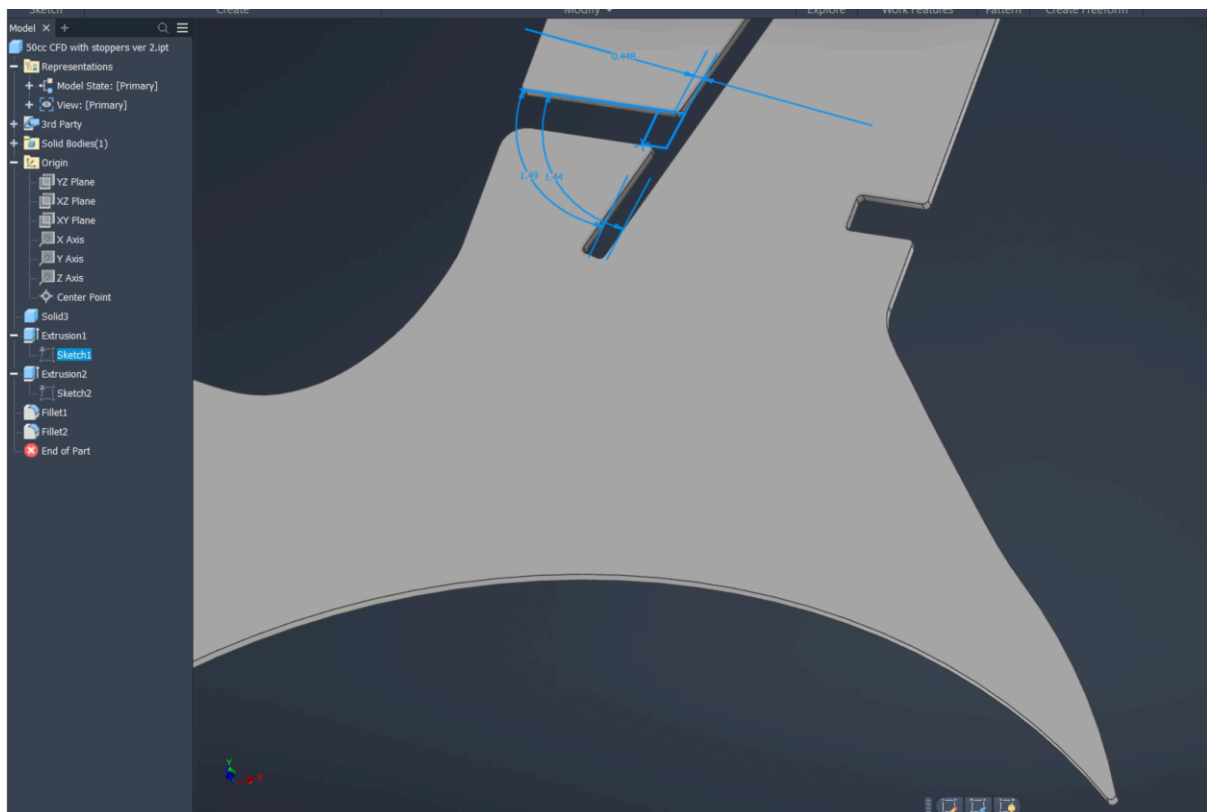
--
VERDICT: Coverage looks healthy - floor is a rare safety
net and transit times span a plausible range.
=====

Yay!

I also checked in haem5_recirculationtest.py if I was turning too many particles to zero velocity (I do that because tiny tiny values in the denominator of my math explodes it), and it seems we aren't. Good. Because stagnant and pressurized areas can be important for raising HI%. Ansys may also be refusing to export recirculation paths amongst my population of pathlines, and that would also cause this issue :(

I realized that meshing is probably something that limits the code's ability to be debugged
I then spent 2 hours refining the mesh poorly. Of the 2D sim . it did not converge

19 Sept 2026



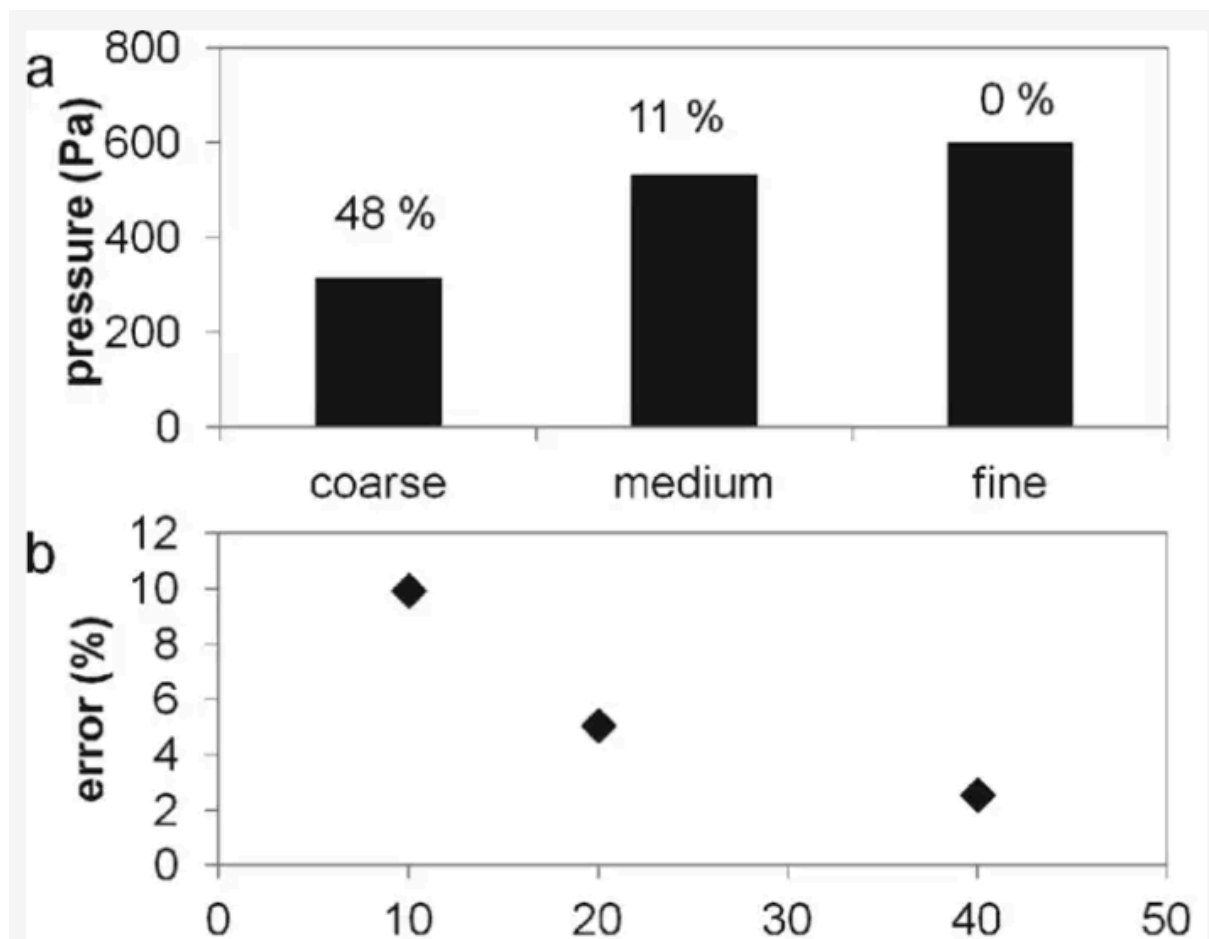
Solving skewness issues resulting in simulations failing by removing the valve gap (above)
and by filletting every edge of the 2D sim - the infinitely fine points of the mesh e.g. 90
degree edges or concentric circle points probably break the solver

The above model uses a common 0.2mm fillet, except for the corners of the diaphragm (0.4mm)

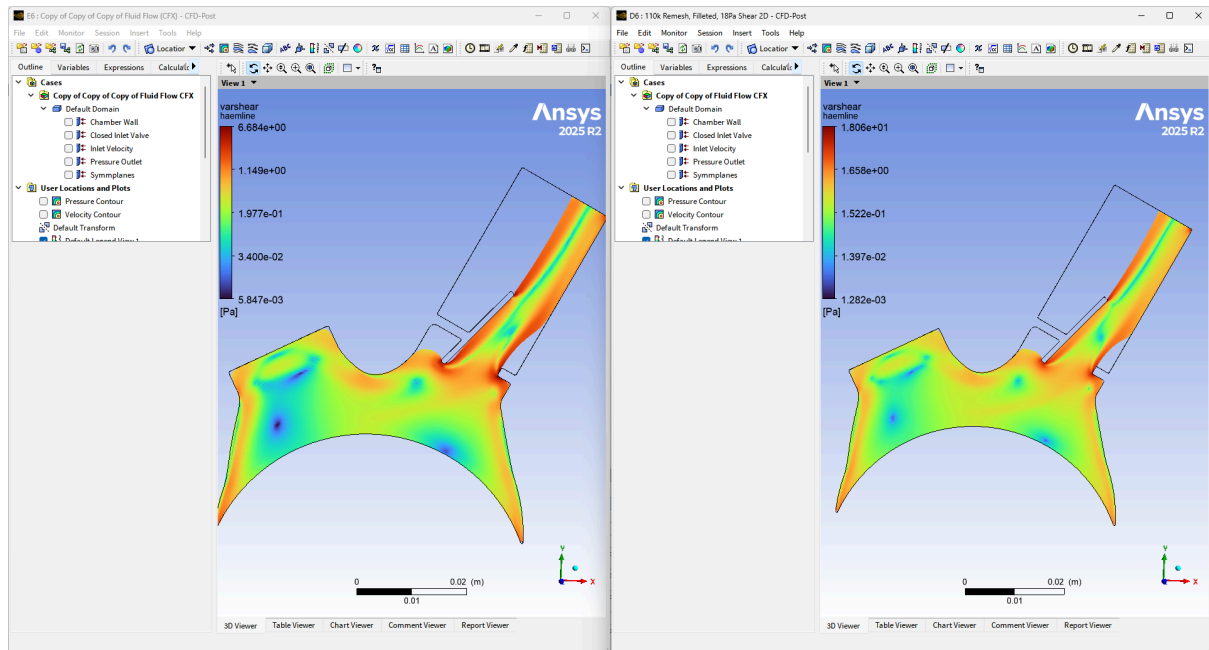
It meshed and simulated !!! YAY !!!!

Result -> 19sept_110kfilleted_18pa_remesh_sevan.wbpz uses a new model (see ipt files for the filleted version uploaded 19 Sept) to smooth all the messy parts of the geometry, which lets me mesh up to 110,000 elements for a simulation time of like ,,, 20 minutes ? for the 2D version at mid systole. It gets a peak velocity of 1.5m/s, a peak internal pressure of ~110mmHg, and the shear is 18 Pa max: 7% haemolysis !!!! Very good !!! Mesh took like 30 seconds to generate :)

asdsa



From blood modelling paper



See S2 Midsem for sept 20 and 21, I forgot to move it here lol

20 Sept 2026 21 Sept 2026

23 Pa from simulating 300k ish elements !!!! I used the initial conditions from the res file of the 110k simulation and it worked instead of breaking !!!

Now trying a 5.6 million element one !!! this has taken 4 hours to process !! (and only 6 minutes to simulate since my ICs were from the 110k mesh)

4:55 PM

evaluation-of-eulerian-an...

Planner x 715 Scribbles x

where Q is the flow rate in the recirculating loop, T is the time the blood has been circulating, and V is the volume of blood in the loop.

The H1h calc we use to study single-pass, vorticity

Results

Hemolyzer-H Model

Mesh refinement. The mesh refinement study was performed with four different meshes of the Hemolyzer-H model, with a refinement ratio of two (Table 2). Convergence in the pressure drop with increasing mesh density was clear (Figure 3A). Meshes were evaluated according to the quantification of uncertainty principles outlined by Roache.^{20,31} The order of convergence was calculated:

$$p = \frac{\log\left(\frac{\phi_{\text{medium}} - \phi_{\text{fine}}}{\phi_{\text{coarse}} - \phi_{\text{fine}}}\right)}{\log(r_r)} \quad (12)$$

where ϕ_{coarse} , ϕ_{medium} , and ϕ_{fine} are the values of the parameter ϕ obtained using coarse, medium, and fine meshes, respectively, and r_r is the refinement ratio (two in our case). ϕ was the area-weighted average of the local shear stresses on the spindle. Meshes one, two, and three were used for the fine, medium, and coarse meshes, respectively. p was calculated as 0.95, showing that the convergence was around first order accurate. The following error estimates were used to calculate the mean percentage errors for each mesh (Figure 3B):

$$E_{\text{fine}} = \frac{|\phi_{\text{medium}} - \phi_{\text{fine}}|}{|\phi_{\text{medium}}|} \cdot \frac{1}{r_r^p - 1} \quad (13)$$

$$E_{\text{medium}} = \frac{|\phi_{\text{coarse}} - \phi_{\text{medium}}|}{|\phi_{\text{medium}}|} \cdot \frac{1}{r_r^p - 1} \quad (14)$$

$$E_{\text{coarse}} = \frac{r_r^p |\phi_{\text{coarse}} - \phi_{\text{medium}}|}{|\phi_{\text{coarse}}|} \cdot \frac{1}{r_r^p - 1} \quad (15)$$

Mesh two was selected for the study, as the error was acceptable (5%), and the mesh size was manageable (<3 million cells).

Taskin et al. Mean Refinement Methodology

→ Refinement Ratio of 2 for 3 mesh densities

'Order of convergence' $p = \frac{\log(\phi_{\text{coarse}} - \phi_{\text{med}})}{\log(r_r)} \cdot \frac{\log(\phi_{\text{med}} - \phi_{\text{fine}})}{\log(r_r)}$

From Roache (20, 31)

$\phi = ?$
↳ Area-weighted avg. shear

→ Then calculate errors from p

$r_r = 1.4$

Mesh: 110 K, 300 K, 5.6 m

↓ verify!

820 K cells

Taskin et al. cites Roach for a mesh error estimation that I can use to figure out if 5.6 million is at ALL acceptable in a pretty cool workflow for assuming error from the diminishing returns end point of a value that is changing with mesh resolution

Next I should:

Double check my refinement ratio from Copy of Copy of Copy of Fluid Flow (E)

Apply it to a new high res sim

Calculate errors from AVERAGE varshear across them 😊

I also need to double check if my hemolysis is ok

<https://claude.ai/chat/0816cab3-36e5-40ed-9264-8d33c832a0bf>

21 Sept 2026

Grid convergence index study !!!!

For 2D filleted mesh simplified

Mesh	Element size	Resulting cells	φ (Pa)
Coarse	0.000175	110,008	18
Medium	0.0001	330,188	24
Fine	0.0000571	998156	25

Taskin et al. equations for p then yield: (see notes screenshot above)

<https://claude.ai/artifact/QR4q7uY5h1ZW2CpeEDo4NG>

$$\varphi = 25 \text{ Pa} \pm 0.6 \text{ Pa} \text{ (GCI}_{\text{fine}} = 2.4 \%, p = 2, F_s = 1.25)$$

Mesh-independent estimate: **25.2–25.5 Pa**. The medium mesh ($h = 0.0001$) is within about 5–6 % of this value.

24 Sept 2026

Double check new code outputs with 330k mesh and 998k mesh

Double check outputs against literature / what I can actually take away from them :)

From the 110k simulation:

```
\Documents\tah-p4p\venv\Scripts\Activate.ps1 py haem5.py export_18pa.csv
```

Hemolysis:

[illegible]

=====

Streamlines processed: 969

Constants used: A=3.62e-05 alpha=2.416 beta=0.785 (GW)

HI2 (temporal-derivative form, Taskin et al. Eq. 4):

Device HI2% (flow-weighted): 4.208324e-06 %

Device NIH (from HI2): 0.4419 mg/100L

HI3 (local/linearized form, Taskin et al. Eq. 5):

Device HI3% (flow-weighted): 7.322561e-06 %

Device NIH (from HI3): 0.7689 mg/100L

SynCardia 50cc literature: 37.15 +/- 12.42 mg/100L

ASTM acceptable limit: 100.0 mg/100L

```

-----
Device SA (median):      0.3372 dyne*s/cm^2
Hellums threshold:      35.0 dyne*s/cm^2
P(SA > Hellums threshold):  0.000 %
Marom et al. medians (ref): {'70cc_TAH': 1.12, '50cc_TAH': 2.19, '35cc_TAH': 3.41}
=====

```

=== Streamline Quality Check ===

```

Total streamlines: 969
Average points per streamline: 109.5954592363261
Min/Max velocity: 0.038 / 1.595 m/s
Min/Max shear: 0.0 / 18.1 Pa
Saved outputs:
  outputs\streamlines_3d_2026-09-24_14-44-56_haem5_export_18pa.png
  outputs\shear_vs_time_2026-09-24_14-44-56_haem5_export_18pa.png
  outputs\hi_histogram_2026-09-24_14-44-56_haem5_export_18pa.png
  outputs\sa_pdf_2026-09-24_14-44-56_haem5_export_18pa.png
  Outputs\streamline_summary_2026-09-24_14-44-56_haem5_export_18pa.csv

```

26 Sept 2026

Code verification with gpt6astra codex medium,

```

1. Document and verify the stress definition

Write the mathematical definition in your methods, alongside the CFX expression. For the usual strain-rate convention,


$$D = \frac{1}{2}(\nabla u + \nabla u^T), \quad \dot{\gamma} = \sqrt{2D:D}, \quad \tau_{\text{scalar}} = \mu \dot{\gamma}$$


Check the convention against your CFX version: ANSYS defines its scalar shear strain rate through a tensor invariant. CFX Theory Guide.

```

https://ansyshelp.ansys.com/public/account/secured?returnurl=/Views/Secured/corp/v242/en/cfx_thry/i1298716.html

FOR REPORT

Do not assume that every paper’s “scalar shear stress” has identical normalization. Compare the expanded expression with Taskin’s Eq. 10, explicitly specifying how tensor components are summed. Different conventions can introduce constant factors that become amplified by the stress exponent.

“Three-dimensional quasi-steady CFD was used to estimate viscous-stress-induced haemolysis during a prescribed mid-systolic operating condition. Flux-weighted predictions were assessed for mesh, trajectory-sampling, and integration sensitivity. Results characterize the specified flow condition and do not represent validated full-cycle haemolysis.”

"Scalar viscous stress was calculated as dynamic viscosity multiplied by the CFX scalar shear strain rate (Eq. 1-49/1-50 in the CFX-Solver Theory Guide). This quantity was verified analytically to be algebraically identical to the scalar shear stress of

Taskin et al. (Eq. 10, based on Bludszuweit's formulation), given the incompressible-flow assumption used in both studies (trace-free viscous stress tensor). Reynolds stresses were excluded from both formulations, consistent with the molecular-viscosity-only scaling of the resolved mean strain rate under SST."

→ outcome here is that the equations that CFX uses and Taskin et al. both are the same for calculating scalar shear stresses

Scalar viscous stress was calculated as dynamic viscosity multiplied by the CFX scalar shear strain rate (CFX-Solver Theory Guide, Eqs. 1-49–1-50). Analytical expansion demonstrated equivalence to Taskin et al.'s scalar stress formulation (Eq. 10) under the incompressible-flow assumption, for which the viscous stress tensor is trace-free. Molecular viscosity and the SST-predicted mean velocity gradients were used; Reynolds stresses were excluded from the haemolysis stress measure, consistent with Taskin et al.

30 Sept 2026

Hemolysis:

100%|
| 985/985 [00:02<00:00, 332.35line/s]

=====
Streamlines processed: 985
Constants used: A=3.62e-05 alpha=2.416 beta=0.785 (GW)

HI2 (temporal-derivative form, Taskin et al. Eq. 4):

Device HI2% (flow-weighted): 1.286683e-04 %

Device NIH (from HI2): 13.5102 mg/100L

HI3 (local/linearized form, Taskin et al. Eq. 5):

Device HI3% (flow-weighted): 2.281012e-04 %

Device NIH (from HI3): 23.9506 mg/100L

SynCardia 50cc literature: 37.15 +/- 12.42 mg/100L

ASTM acceptable limit: 100.0 mg/100L

Device SA (median): 0.6433 dyne*s/cm^2

Hellums threshold: 35.0 dyne*s/cm^2

P(SA > Hellums threshold): 0.000 %

Marom et al. medians (ref): {'70cc_TAH': 1.12, '50cc_TAH': 2.19, '35cc_TAH': 3.41}

=====
=== Streamline Quality Check ===

Total streamlines: 985

Average points per streamline: 91.95228426395938

Min/Max velocity: 0.019 / 3.379 m/s

Min/Max shear: 0.0 / 94.4 Pa

Saved outputs:

outputs\streamlines_3d_2026-09-30_21-00-58_haem5_1000p_10ks_alotpa_3d.png

N_requested	n_streamlines	constants	HI2_percent	HI3_percent	HI2_NIH_mg_per_100L	HI3_NIH_mg_per_100L
10	10	GW	1.743e-05	3.345e-05	1.83	3.512
30	30	GW	6.623e-05	0.0001532	6.955	16.09
100	100	GW	8.498e-05	0.0001833	8.923	19.24
300	298	GW	0.0001232	0.0002143	12.94	22.5
1000	985	GW	0.0001287	0.0002281	13.51	23.95
3000	2905	GW	0.0001374	0.0002343	14.42	24.6
10000	9709	GW	0.0001638	0.0002738	17.2	28.75
30000	29451	GW	0.0001745	0.0002839	18.32	29.81
10	10	HO	4.387e-07	7.747e-07	0.04607	0.08135
30	30	HO	1.094e-06	2.283e-06	0.1148	0.2397
100	100	HO	1.334e-06	2.677e-06	0.1401	0.2811
300	298	HO	1.936e-06	3.262e-06	0.2033	0.3425
1000	985	HO	2.038e-06	3.444e-06	0.214	0.3616
3000	2905	HO	2.197e-06	3.571e-06	0.2306	0.3749
10000	9709	HO	2.508e-06	4.017e-06	0.2633	0.4218
30000	29451	HO	2.649e-06	4.146e-06	0.2781	0.4353
10	10	TZ	3.676e-06	8.944e-06	0.386	0.9391
30	30	TZ	8.614e-06	2.673e-05	0.9044	2.806
100	100	TZ	1.081e-05	3.119e-05	1.135	3.275
300	298	TZ	1.577e-05	3.491e-05	1.655	3.665
1000	985	TZ	1.696e-05	3.789e-05	1.78	3.978
3000	2905	TZ	1.838e-05	3.88e-05	1.93	4.074
10000	9709	TZ	2.076e-05	4.298e-05	2.179	4.512
30000	29451	TZ	2.192e-05	4.392e-05	2.302	4.611

(.venv) PS C:\Users\kingk\Documents\tah-p4p> python plot_sweep.py

HI2-GW (reference = finest level, 29451 lines)

lines	value	% vs finest	% vs previous
10	1.743e-05	-90.01	NaN
30	6.623e-05	-62.04	280
100	8.498e-05	-51.3	28.3
298	0.0001232	-29.38	45.01
985	0.0001287	-26.26	4.413
2905	0.0001374	-21.28	6.753
9709	0.0001638	-6.136	19.24
29451	0.0001745	0	6.538

HI3-GW (reference = finest level, 29451 lines)

lines	value	% vs finest	% vs previous
10	3.345e-05	-88.22	NaN
30	0.0001532	-46.02	358.2
100	0.0001833	-35.45	19.59
298	0.0002143	-24.53	16.92
985	0.0002281	-19.65	6.454
2905	0.0002343	-17.47	2.724
9709	0.0002738	-3.542	16.87
29451	0.0002839	0	3.672

HI2-HO (reference = finest level, 29451 lines)

lines	value	% vs finest	% vs previous
10	4.387e-07	-83.43	NaN
30	1.094e-06	-58.71	149.3
100	1.334e-06	-49.63	21.98
298	1.936e-06	-26.9	45.13
985	2.038e-06	-23.05	5.271
2905	2.197e-06	-17.07	7.77
9709	2.508e-06	-5.327	14.16
29451	2.649e-06	0	5.627

HI3-HO (reference = finest level, 29451 lines)

lines	value	% vs finest	% vs previous
10	7.747e-07	-81.31	NaN
30	2.283e-06	-44.94	194.7
100	2.677e-06	-35.42	17.28
298	3.262e-06	-21.32	21.83
985	3.444e-06	-16.93	5.59
2905	3.571e-06	-13.87	3.675
9709	4.017e-06	-3.116	12.49
29451	4.146e-06	0	3.216

HI2-TZ (reference = finest level, 29451 lines)

lines	value	% vs finest	% vs previous
-------	-------	-------------	---------------

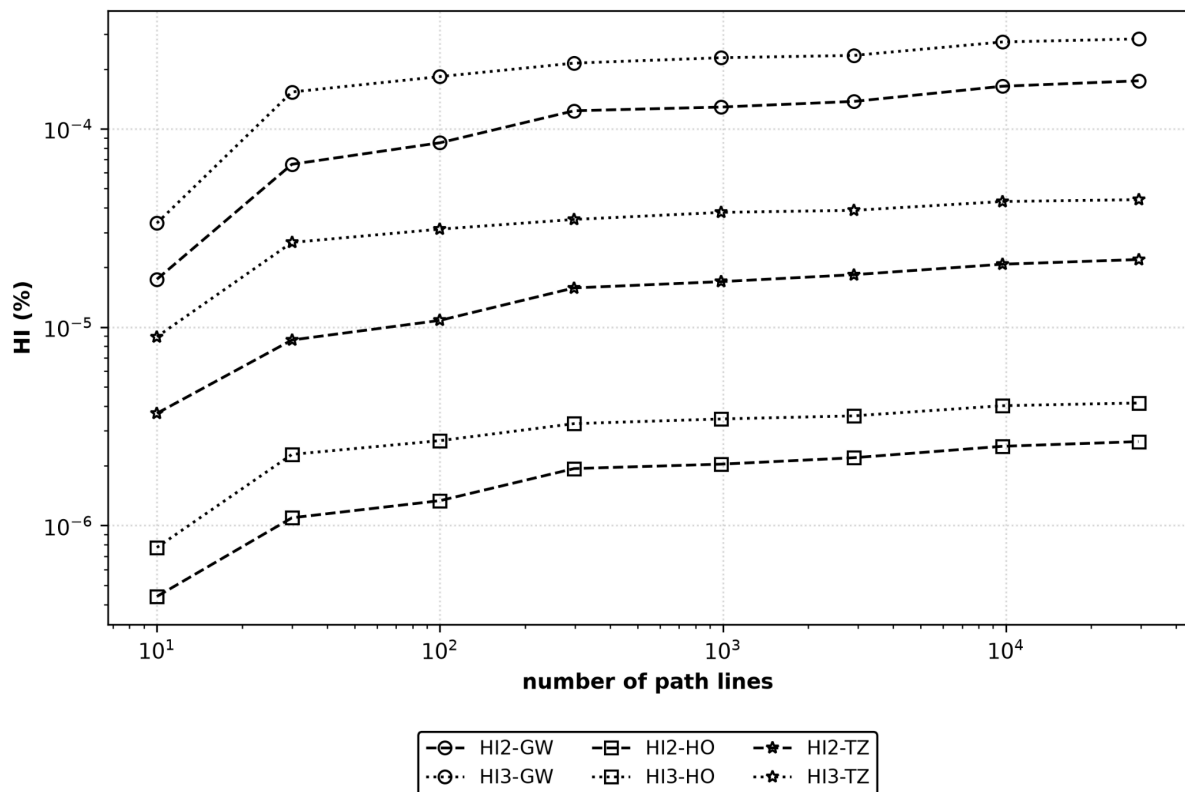
10	3.676e-06	-83.23	NaN
30	8.614e-06	-60.7	134.3
100	1.081e-05	-50.67	25.53
298	1.577e-05	-28.08	45.8
985	1.696e-05	-22.65	7.549
2905	1.838e-05	-16.13	8.422
9709	2.076e-05	-5.312	12.9
29451	2.192e-05	0	5.61

HI3-TZ (reference = finest level, 29451 lines)

lines	value	% vs finest	% vs previous
10	8.944e-06	-79.64	NaN
30	2.673e-05	-39.15	198.8
100	3.119e-05	-28.98	16.7
298	3.491e-05	-20.52	11.92
985	3.789e-05	-13.74	8.536
2905	3.88e-05	-11.66	2.406
9709	4.298e-05	-2.149	10.77
29451	4.392e-05	0	2.196

Saved hi_vs_pathlines.png

(.venv) PS C:\Users\kingk\Documents\tah-p4p>



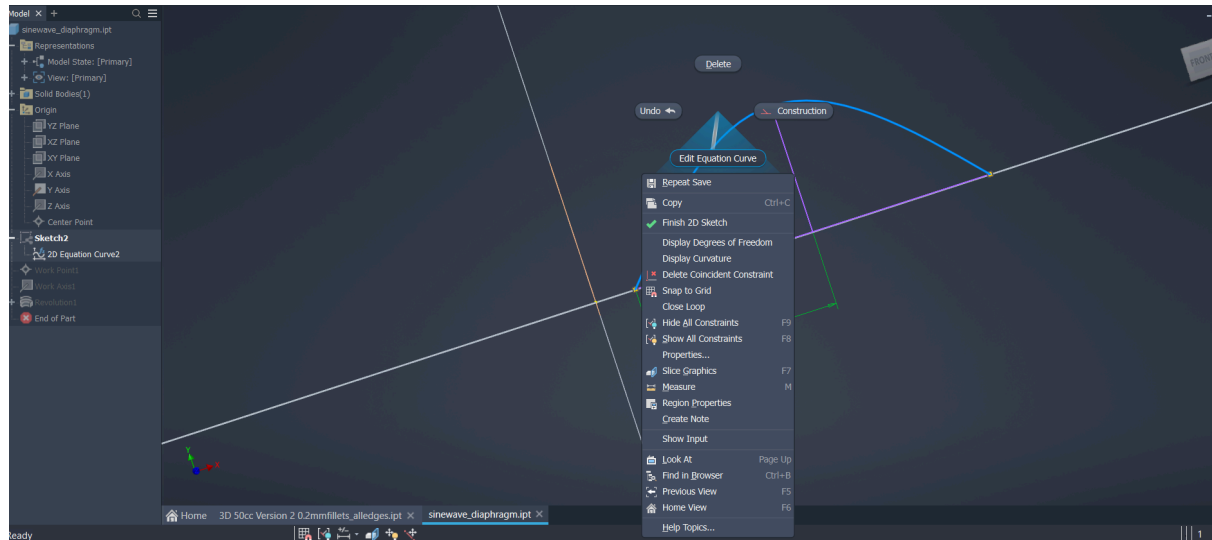
Todo next time : random point testing instead of equally spaced

Test step file visualization

4 Oct 2026

Does residual amount $e-3$ vs $e-5$ affect hemolysis ??

Can I drive a geometry boundary in Inventor with a sine wave?



Yes

Whats happening here is that the original TAH model no longer has a diaphragm, a separate file is revolving a parametric equation to make a dome, and an assembly is bringing those together to make a new ipt that booleans the dome out of the TAH model fluid domain

Pump : 1 is 20mm rise, 0.001 is almost 0, 0.5 is 10mm, it is an analogue for the diaphragm

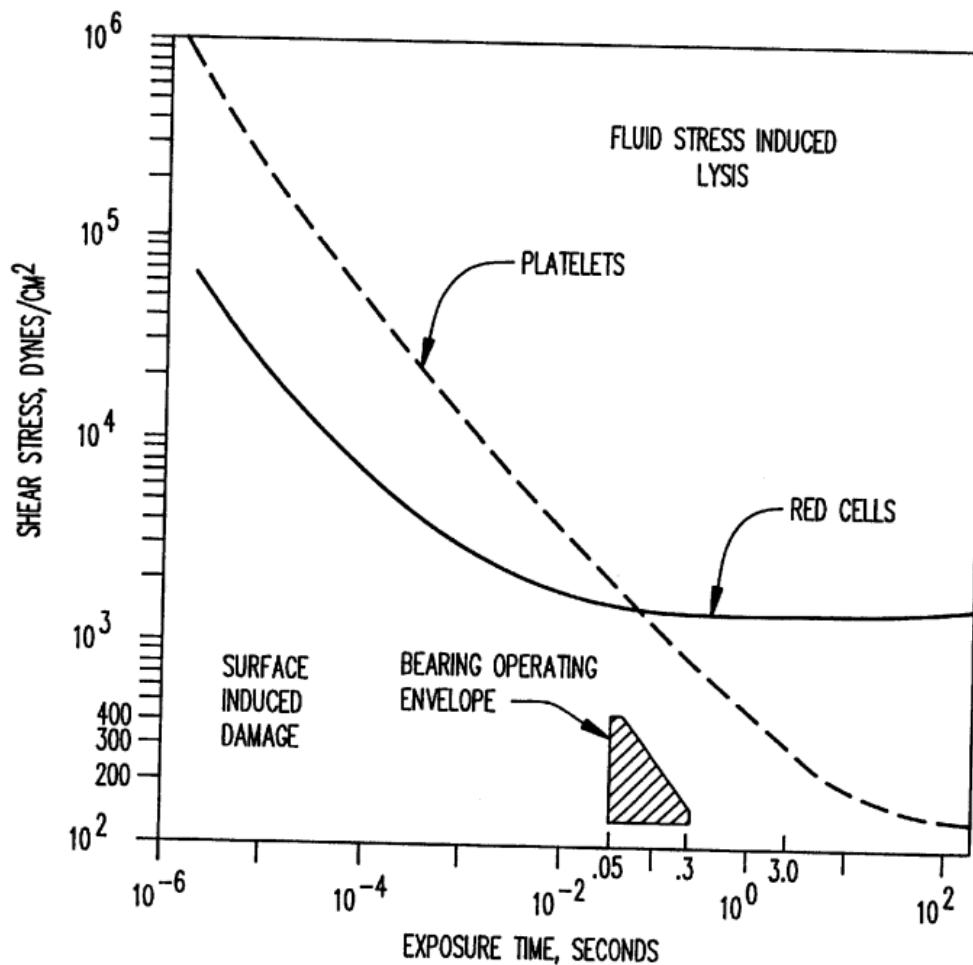


FIG. 26

<https://patentimages.storage.googleapis.com/57/c0/87/b31a50ab08f215/US5924975.pdf>
 US5924975A - Michael P. Goldowsky - Linear pump

5 Oct 2026

<https://chatgpt.com/c/6ac2e184-5524-83ec-9936-37dc0c572346>

Based on the patent above, made a threshold checker to compliment the mathematical modelling of hemolysis - there's gonna be biological thresholds where things just blow up regardless of the accumulated time / stress

@Hemolysis does residual amount e-3 vs e-5 affect hemolysis ??

if stable for flow rate and pressure / shear stress then we can just ASSUME the solution is verified / stable for using in the haem calcs

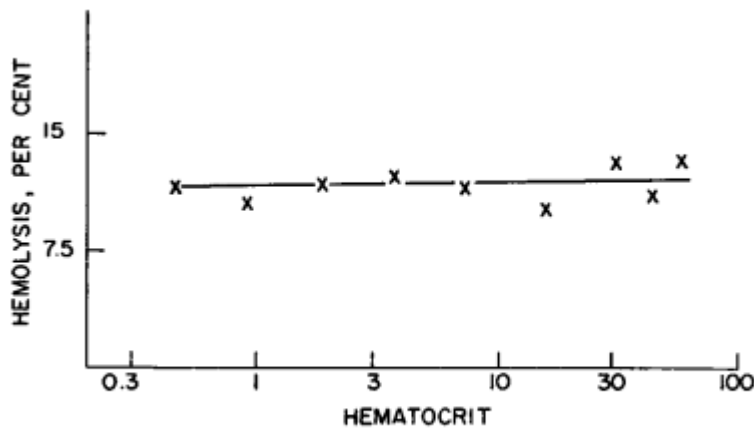


FIGURE 5 Hemolysis at various values of the hematocrit.

[https://www.cell.com/biophysj/pdf/S0006-3495\(72\)86085-5.pdf?_returnURL=https%3A%2F%2Flinkinghub.elsevier.com%2Fretrieve%2Fpii%2FS0006349572860855%3Fshowall%3Dtrue](https://www.cell.com/biophysj/pdf/S0006-3495(72)86085-5.pdf?_returnURL=https%3A%2F%2Flinkinghub.elsevier.com%2Fretrieve%2Fpii%2FS0006349572860855%3Fshowall%3Dtrue)

Hematocrit concentration and resulting hemolysis percentagewise is not correlated

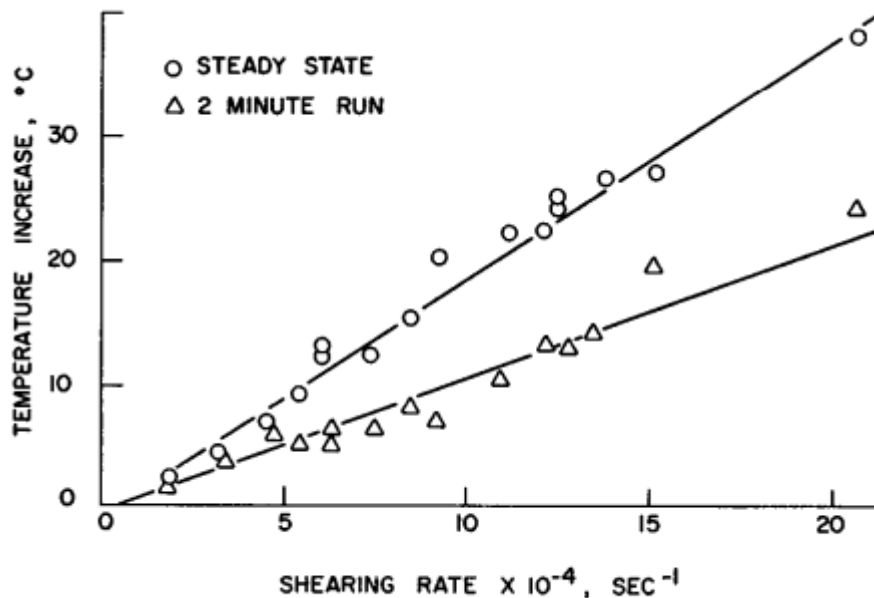


FIGURE 6 Temperature increase due to viscous heating.

Can mention this for future recommendations (temperature is concerning)

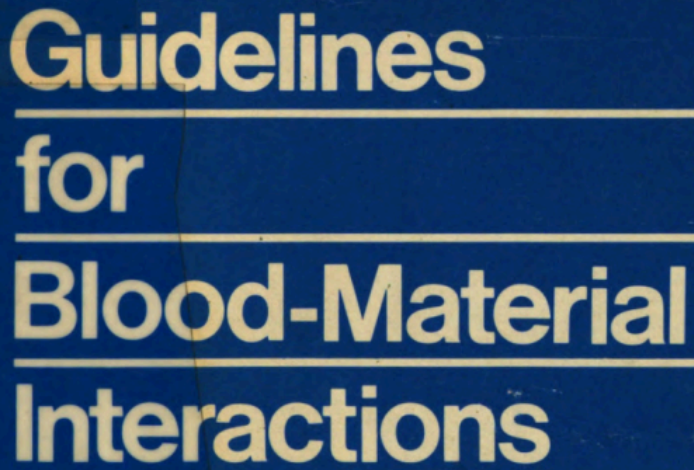
RED BLOOD CELL DAMAGE BY SHEAR STRESS

L. B. LEVERETT, J. D. HELLUMS, C. P. ALFREY, and E. C. LYNCH

From the Biomedical Engineering Laboratory, Rice University, Houston, Texas 77025, and Baylor University College of Medicine, Houston, Texas 77025

<https://chatgpt.com/c/6ac2e184-5524-83ec-9936-37dc0c572346>

file:///C:/Users/kingk/Downloads/Guidelines_for_Blood_material_Interactio.pdf



Report of the National Heart, Lung, and Blood Institute Working Group

criteria	max_utilization	streamlines_unassessed	streamlines_exceeding_percent	flow_weighted_exceedance_percent	worst_stress_Pa	worst_contiguous_duration_s	status
----------	-----------------	------------------------	-------------------------------	----------------------------------	-----------------	-----------------------------	--------

Red blood cells	0.1845	0	0	0
39.93	0.004076 BELOW LYSIS REFERENCE			
Platelets	0.07072	0	0	0
36.74	0.005075 BELOW LYSIS REFERENCE			

Interpretation: utilization = actual stress / empirical critical stress

utilization < 1 -> below empirical boundary

utilization >= 1 -> boundary crossed

NOTE: Approximate NIH Fig. 3.6 traces; screening for lysis, not activation.

Repeated separated exposures are not accumulated by this envelope.

=====

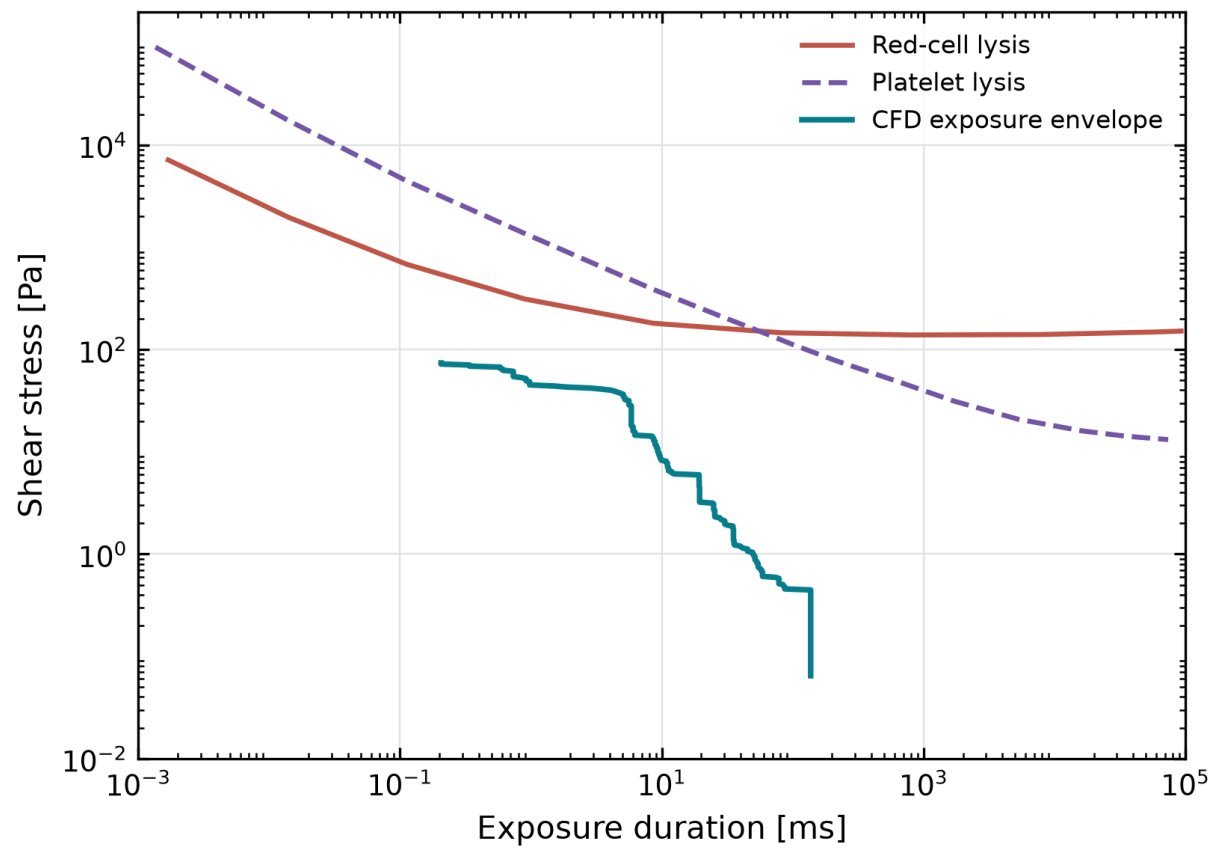
Saved:

outputs_empirical\empirical_threshold_summary_2026-10-05_20-53-32_1000p_100_alotpa_3djawn.csv

outputs_empirical\empirical_threshold_device_2026-10-05_20-53-32_1000p_100_alotpa_3djawn.csv

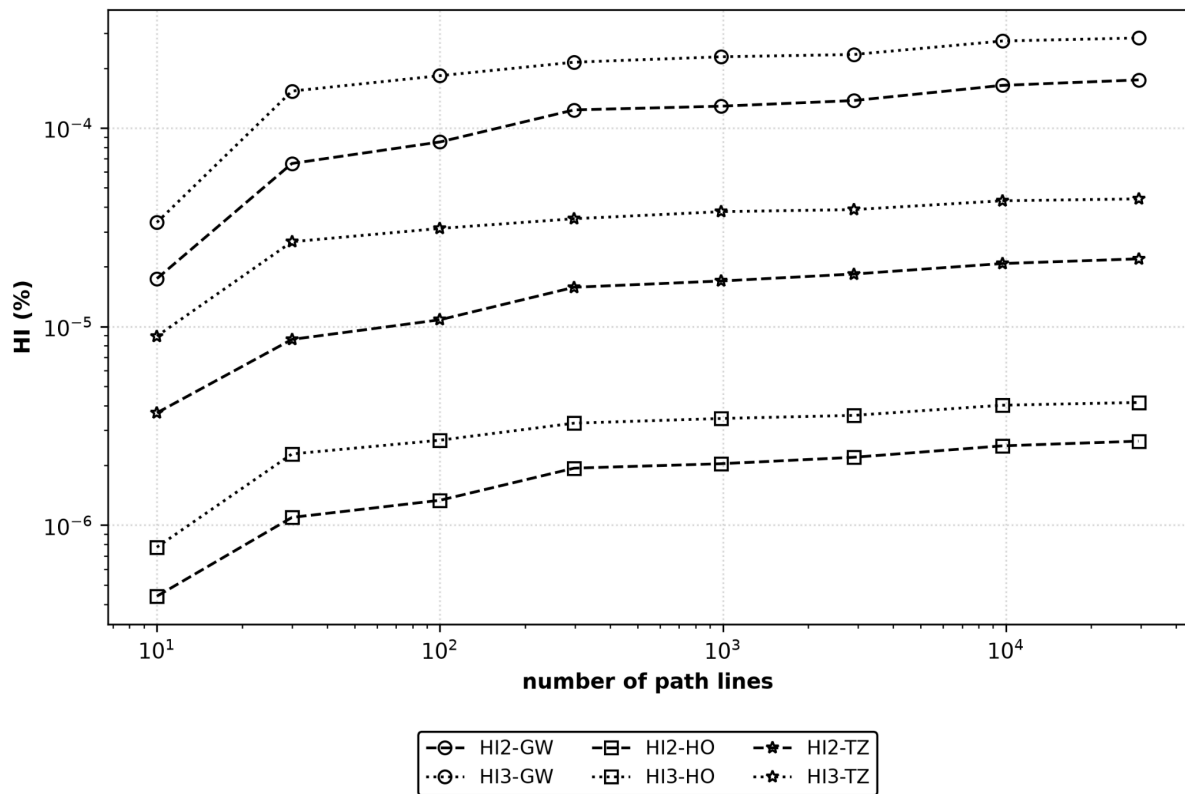
outputs_empirical\empirical_exposure_envelope_2026-10-05_20-53-32_1000p_100_alotpa_3djawn.csv

outputs_empirical\empirical_stress_duration_2026-10-05_20-53-32_1000p_100_alotpa_3djawn.png



Empirical_stress_duration_report_ms_1000p_100_alotpa_3djawn.png (above) got with
python haem5_empiricalthresholds.py 1000p_100_alotpa_3djawn.csv

ALSO RE: THE SWEEP



For your sweep, use:

$$\delta_N = \frac{|HI_N - HI_{\text{ref}}|}{|HI_{\text{ref}}|} \times 100\%$$

Here, initially take HI_{ref} from 30,000 requested pathlines—29,451 actually processed. This is a *provisional reference*, since the last point is not automatically the converged solution.

Calculated from your CSV, the differences relative to that reference are: [sweep_results.csv](#)

<https://chatgpt.com/c/6ac35d45-8e50-83ec-a0ca-58ab2fd88bc0>

WE CAN JUSTIFY our choice of pathline count by assuming 30k as a ‘reference’ or an assumed convergence, because we have nothing to compare to. It just seems to change my diminishing amounts from then on despite exp. Increasing processing time.

Requested pathlines	HI2-GW	HI3-GW	HI2-HO	HI3-HO	HI2-TZ	HI3-TZ
1,000	26.26%	19.65%	23.05%	16.93%	22.65%	13.74%
3,000	21.28%	17.47%	17.07%	13.87%	16.13%	11.66%
10,000	6.14%	3.54%	5.33%	3.12%	5.31%	2.15%

Via sweep results (NOTE THAT TZ IS IRRELEVANT TO OUR INVESTIGATION)

```
##### N = 100000 (sweep_N100000.csv) #####
```

100%|

[GW] fitted range: $\tau < 255 \text{ Pa}$, $t < 700 \text{ ms}$

Hemolysis:

97331/97331 [05:31<00:00, 293.45line/s]

Constant-set comparison (device-level, flow-weighted):

GW	0.0001617	0.0002676	16.98	28.1	0
	0.0002226				

done in 1083.2 s, saved C:\Users\kingk\Documents\tah-p4p\sweeps\sweep_results.csv

100	100	GW	8.498e-05	0.0001833	8.923	19.24
300	298	GW	0.0001232	0.0002143	12.94	22.5
1000	985	GW	0.0001287	0.0002281	13.51	23.95
3000	2905	GW	0.0001374	0.0002343	14.42	24.6
10000	9709	GW	0.0001638	0.0002738	17.2	28.75
30000	29451	GW	0.0001745	0.0002839	18.32	29.81
100000	97331	GW	0.0001617	0.0002676	16.98	28.1
10	10	HO	4.387e-07	7.747e-07	0.04607	0.08135
30	30	HO	1.094e-06	2.283e-06	0.1148	0.2397
100	100	HO	1.334e-06	2.677e-06	0.1401	0.2811
300	298	HO	1.936e-06	3.262e-06	0.2033	0.3425
1000	985	HO	2.038e-06	3.444e-06	0.214	0.3616
3000	2905	HO	2.197e-06	3.571e-06	0.2306	0.3749
10000	9709	HO	2.508e-06	4.017e-06	0.2633	0.4218
30000	29451	HO	2.649e-06	4.146e-06	0.2781	0.4353
100000	97331	HO	2.477e-06	3.934e-06	0.26	0.4131
10	10	TZ	3.676e-06	8.944e-06	0.386	0.9391
30	30	TZ	8.614e-06	2.673e-05	0.9044	2.806
100	100	TZ	1.081e-05	3.119e-05	1.135	3.275
300	298	TZ	1.577e-05	3.491e-05	1.655	3.665
1000	985	TZ	1.696e-05	3.789e-05	1.78	3.978
3000	2905	TZ	1.838e-05	3.88e-05	1.93	4.074
10000	9709	TZ	2.076e-05	4.298e-05	2.179	4.512
30000	29451	TZ	2.192e-05	4.392e-05	2.302	4.611

=====

We need a segment count sweep 😞

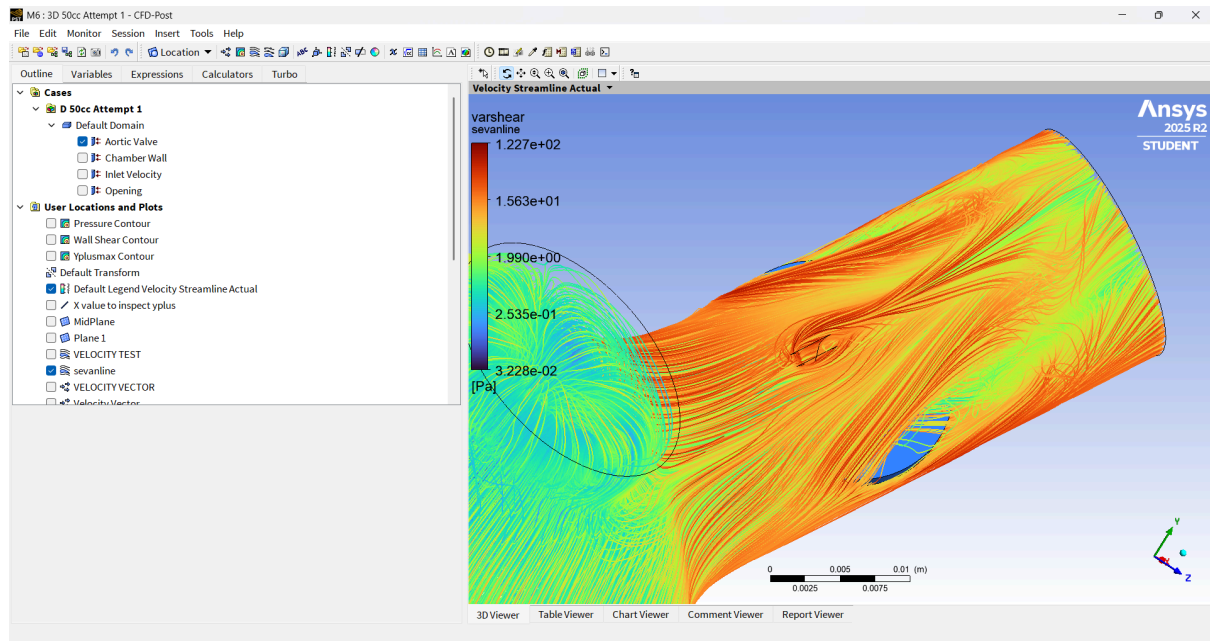
And a bigger point sweep / ~~random seeding~~ random seeding is impossible

<https://pmc.ncbi.nlm.nih.gov/articles/PMC4854792/>

ALSO READ THIS

Because our data has like <1ms points of huge shear : like up to 75 Pa or greater, which for repeated systole cycles this can be genuinely important for cumulative damage

NAH



So pretty

6 Oct 2026

We may need to consider sensitization in the hemolysis code `P4P SCRIBBLES`, as brief spikes of many dynes can be bad for platelets even though we don't maintain those values

I'm getting the code to look into that now

Currently a 1M node hemolysis calculation is running to find the 'convergence' of the line count, where after I can find a line count that I can use for all future tests. Following this I will investigate what segment count is needed.

Volume Flow Rate Calculation

We make a new expression in CFD-Post with definition

`area()@Opening * areaAve(Velocity)@Opening`

For 3D 50cc Attempt 1: $0.000472695 \text{ [m}^3 \text{ s}^{-1}\text{]}$

Technically wrong, this is better : `areaInt( Velocity u * Normal X + Velocity v * Normal Y + Velocity w * Normal Z)@Opening`

Yields about 28 litres per minute !!!!!!!!! OH !!!!!!!!! OH NO !!!!!!!!!

<https://chatgpt.com/c/6ac4818d-f788-83ec-84d7-d098ef3ead5b>

IMPORTANT !

Yes. For a quasi-steady CFD model sampled at 4–5 points through systole, the cleanest way to estimate cardiac output is:

$$\text{CFD } Q(t_i) \rightarrow \text{integrate over systole} \rightarrow SV \rightarrow CO$$

You should **not** use the peak flow rate, and I would not simply average the 4–5 values unless the time spacing is uniform and you explicitly account for systolic duration.

For every quasi-steady state, define one CFX-Post expression:

</> Plain text

```
Qout =  
areaInt(  
    Velocity u * Normal X  
    + Velocity v * Normal Y  
    + Velocity w * Normal Z  
)@Opening
```

That gives Q_i in m^3/s for each systolic time point.

Suppose your cases represent:

Systolic time	CFD flow
t_0	Q_0
t_1	Q_1
t_2	Q_2
t_3	Q_3
t_4	Q_4

Then estimate the ejected stroke volume with trapezoidal integration:

$$SV \approx \sum_{i=0}^{n-1} \frac{Q_i + Q_{i+1}}{2} (t_{i+1} - t_i)$$

Because Q is in m^3/s and t is seconds, the result is m^3/beat .

Then:

$$SV_{\text{mL}} = \downarrow SV_{\text{m}^3} \times 10^6$$

and finally:

$$CO = \frac{SV_{mL} \times HR}{1000}$$

in L/min.

So here are the 5 Systolic phases to test

0%

25%

50%

75%

100%

VALIDATION !!!!!

There's also a useful independent sanity check for your TAH: if your chamber volume change over systole is known from the geometry/motion, then ideally

$$SV_{CFD} = \int Q dt \approx \Delta V_{chamber}$$

assuming negligible leakage/compressibility. That is an excellent validation check for the quasi-steady workflow.

Validation checklist:

Make sure the CO is physiological, internal pressure avg/max is, etc

Pathline / Segment Analysis Continued

<https://chatgpt.com/c/6ac35d45-8e50-83ec-a0ca-58ab2fd88bc0>

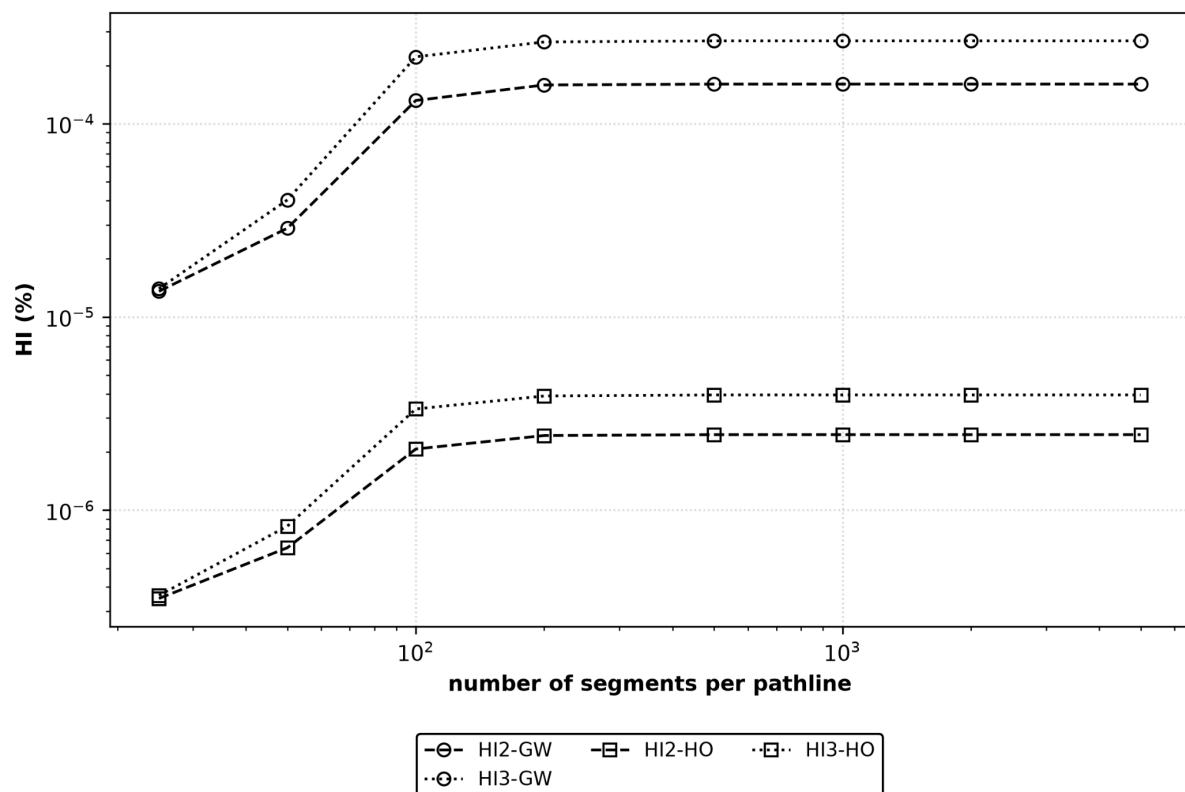
Requested pathlines	HI2-GW	HI3-GW	HI2-HO	HI3-HO	All ≤5%?
1,000	20.64%	14.80%	17.87%	12.49%	No
2,000	0.09%	3.25%	3.82%	0.55%	Yes
3,000	15.28%	12.48%	11.49%	9.28%	No

4,000	4.56%	4.52%	2.80%	2.49%	Yes
5,000	1.10%	0.25%	0.89%	0.30%	Yes
6,000	5.10%	4.33%	2.74%	2.35%	No
7,000	6.16%	6.11%	5.41%	5.34%	No
8,000	0.47%	0.16%	1.24%	0.64%	Yes
9,000	6.53%	7.05%	5.10%	5.56%	No
10,000	1.02%	2.28%	1.05%	2.06%	Yes

Eventually

Check the final combination on the 35cc model by comparing 5k against a larger count, such as 20k. If the comparison fails, increase the count. Agreement is a practical sensitivity check, not a guaranteed error bound.

Asdpoadsp for pathlines noadniadonid



NEXT : do a step tolerance sweep

7 Oct 2026

6 lite

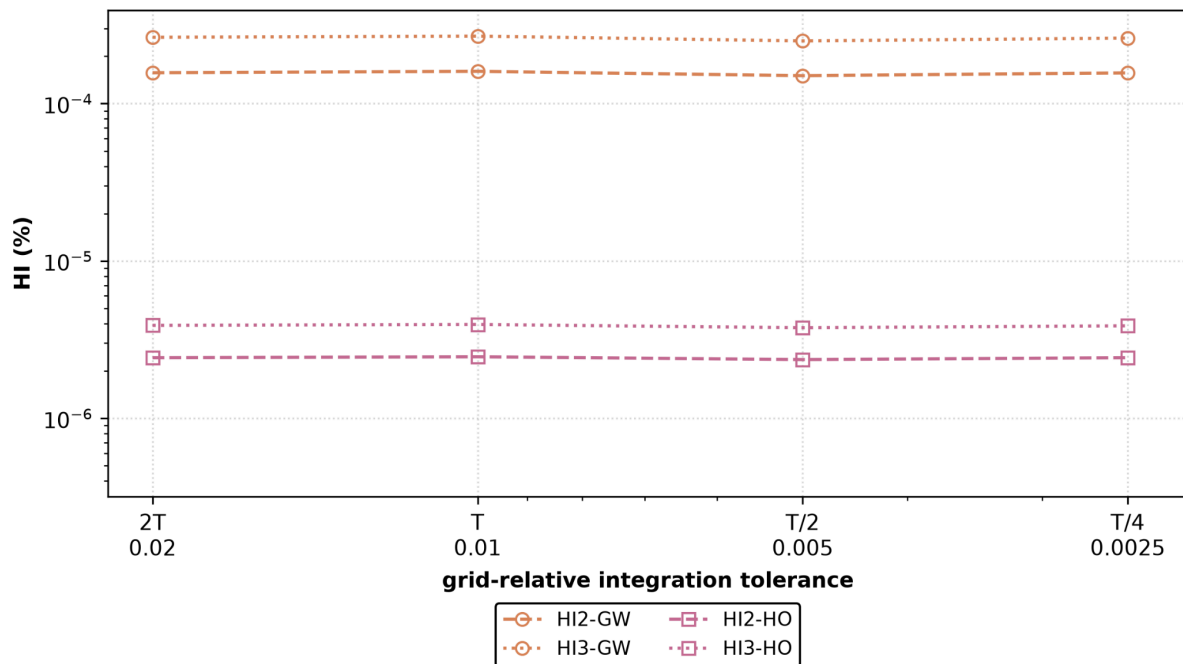
**VERY GOOD: MATH FOR EJECTED-VOLUME WEIGHTED HEMOLYSIS SUM !!!
FOR WHOLE CYCLE HEMOLYSIS**

If calculating a combined snapshot result, use ejected-volume weighting:

$$HI_{QS} = \frac{\sum_k (HI_k * Q_k * \Delta t_k)}{\sum_k (Q_k * \Delta t_k)}$$

In investigation plan markdown file

Tolerance_sweep_results.csv



Tolerance has an on average <5% variation from T, and the computation time barely changes so I'm gonna leave it on the ANSYS default of 0.01

For your P4P, I think the defensible architecture is therefore

You establish a prescribed TAH stroke:

$$V(t)$$

from your diaphragm motion.

From that you obtain:

$$Q(t) = -\dot{V}(t).$$

Then choose your four representative systolic phases:

$$t_1, t_2, t_3, t_4.$$

For each:

$$t_i \rightarrow [V_i, Q_i, \text{geometry}_i] \rightarrow \text{steady CFX}$$

CFX then gives you:

velocity distribution, τ , pathlines, haemolysis metrics, ...

And separately:

$$SV = V_{ED} - V_{ES}$$

$$CO = SV \times HR$$

So **CO** is not an output you're trying to discover from the four quasi-steady CFD simulations. It is an operating parameter implied by the TAH motion you're modelling.

Your 12.6 cm/s number is actually correct as the **maximum** velocity under your assumptions. At 120 bpm,

$$T = \frac{60}{120} = 0.5 \text{ s}$$

and if systolic ejection occupies half the cycle,

$$T_s = 0.25 \text{ s.}$$

For a total diaphragm travel of

$$S = 0.02 \text{ m,}$$

a convenient displacement law over systole is

$$x(t) = \frac{S}{2} \left[1 - \cos \left(\frac{\pi t}{T_s} \right) \right], \quad 0 \leq t \leq T_s.$$

Then the actual diaphragm velocity is

$$v(t) = \frac{\pi S}{2T_s} \sin \left(\frac{\pi t}{T_s} \right)$$

and therefore

$$v_{\max} = \frac{\pi(0.02)}{2(0.25)} = 0.1257 \text{ m/s} = \boxed{12.57 \text{ cm/s}}.$$

^ solution for diaphragm velocity at any given t_n

For each t , we:

Systolic phase	Position through 20 mm stroke	Diaphragm velocity
20%	1.91 mm	7.39 cm/s
40%	6.91 mm	11.95 cm/s
60%	13.09 mm	11.95 cm/s
80%	18.09 mm	7.39 cm/s

<https://chatgpt.com/share/6ac5f562-9118-83ec-ae35-4d91df3dfa4b>

^ public link for this conversation

Case 1

systolic phase = 20%

diaphragm geometry = $x(0.20)$

diaphragm velocity = $v(0.20)$

Case 2

systolic phase = 40%

diaphragm geometry = $x(0.40)$

diaphragm velocity = $v(0.40)$

Case 3

systolic phase = 60%

diaphragm geometry = $x(0.60)$

diaphragm velocity = $v(0.60)$

Case 4

systolic phase = 80%

diaphragm geometry = $x(0.80)$

diaphragm velocity = $v(0.80)$

1. Hemolysis Quantification is Superficial and Indirect

This is probably your biggest and most exploitable gap.

The paper uses **wall shear stress (Pa) in a simplified tube** as a proxy for hemolysis risk, with a binary pass/fail against 150 Pa. This misses almost everything about how hemolysis actually works in a TAH:

- No **Hemolysis Index (HI)** is ever calculated. The standard approach uses the Giersiepen-Wurzinger power law model: $HI\% = C \times \tau^\alpha \times t^\beta$, where τ is shear stress and t is exposure time. The paper never applies this.
- No conversion to **mg/100 L** or **g/100 L**, so there's no comparison possible against the NIH threshold (>0.1 g/100 L indicating hemolysis) or against the clinically measured 37.15 ± 12.42 mg/100 L from the 50cc SynCardia — which you already have from your meeting notes. That's a glaring gap.
- Their 2% shear stress criterion (comparing 35cc to 50cc outlet tube) **failed**, and they simply accepted this because the other criterion passed. There's no discussion of what that failure actually means physiologically.

Your gap to exploit: Compute a proper HI and convert to mg/100 L for direct clinical comparison using your CFD data from the actual chamber geometry, not a tube.

2. CFD is on a Tube, Not the Ventricle Geometry

The CFD analysis was not performed on the 3D ventricular model — it was done on **two separate simplified cylindrical tubes** representing only the outlet channel. This is a fundamental limitation:

- No analysis of **internal flow patterns, recirculation zones, or stagnation regions** inside the blood chamber, which are primary sites of hemolysis and thrombosis in pulsatile TAHs.
- The **diaphragm dynamics** (which drive ejection and filling) are completely absent from the fluid analysis. The moving boundary interaction with blood is a known driver of hemolysis in pneumatic TAHs.
- No **valve hemodynamics** — mechanical heart valves (tilting disc or bileaflet) are major hemolysis and thrombosis hotspots, and the paper doesn't address them at all.
- **Steady-state flow was assumed.** A TAH is inherently pulsatile — steady CFD misses peak shear stress events during valve opening/closing and rapid ejection phases.

Your gap: Running CFD on the actual 3D chamber with pulsatile boundary conditions and moving-mesh or dynamic diaphragm modelling gives you results this paper cannot produce.

3. Two of the Four Engineering Requirements Were Never Verified

The paper explicitly states ejection fraction and cardiac output verifications **were not completed**. This means:

- There is **no demonstrated evidence** that the 35cc design produces a cardiac output in the 2.6–5.4 L/min range for the target age group.
- There is **no demonstrated ejection fraction ≥ 0.61** .
- The design is declared "feasible" based on only two of four requirements — size and a partial hemolysis pass.

Your gap: Your flow loop directly measures cardiac output, and your CFD can estimate ejection fraction. You will be the first to actually validate these for a sub-50cc TAH-like device.

4. Cardiac Output is Not Normalized for Pediatric Patients

The paper cites a cardiac output target of **2.6–5.4 L/min** (from Wu et al. 2016), which covers a wide age range. For your target population (6–10 year olds):

- **Cardiac index** (L/min/m²) is the appropriate normalized metric for pediatric patients and varies significantly with BSA. A 6-year-old and a 10-year-old have meaningfully different requirements.
- Stroke volume is never explicitly verified either — your meeting notes correctly identify 30–40 mL as a physiologically grounded target, but this paper doesn't state a stroke volume target at all.
- Mean arterial pressure targets are also absent from the verification, despite being a direct output of any real flow loop.

Your gap: Design your loop to target cardiac index per BSA rather than absolute cardiac output, and measure MAP as a primary outcome.

5. No Physical Flow Loop or Mock Circulatory Loop Validation

The paper produces a **non-operational 3D print** and software-only verification. There is no:

- Mock circulatory loop (MCL) testing with blood analogue fluid
- Pressure–volume loop characterisation
- Measurement of stroke volume under realistic afterload/preload conditions
- Frank–Starling behaviour assessment (which they mention as important but never test)

This is a textbook gap — computational predictions without experimental validation are insufficient for medical device development, and the paper acknowledges this implicitly.

Your gap: Your physical flow loop is direct experimental evidence that no version of this work has ever produced.

6. Right Ventricle is Entirely Unmodelled

The paper only models the left ventricle, with the stated assumption that the right ventricle is "similar in design" and simply doubles the external volume. In reality:

- Right and left ventricular pressures differ significantly (pulmonary ~25 mmHg systolic vs systemic ~120 mmHg)
- A properly functioning TAH must balance output between left and right to avoid pulmonary oedema or systemic congestion
- The paper's 2–5% reduced right output noted in your meeting notes is clinically relevant and never addressed

Your gap: Even a simplified two-chamber loop that demonstrates ventricular balance (or quantifies imbalance) would be a novel contribution.

7. No Thrombosis Risk Analysis

Hemolysis and thrombosis are linked — regions of high shear cause haemolysis, while regions of **low shear and stagnation** cause thrombus formation. The paper looks only at peak shear stress and ignores:

- Residence time of blood in the chamber
- Recirculation regions near inlet/outlet junctions
- Valve wake regions

These are addressable with CFD post-processing (e.g. particle tracking, residence time distribution).

Summary Table

Gap	Paper Status	Your Approach
Hemolysis Index (mg/100 L)	Not calculated	CFD power law + compare to 50cc clinical data
CFD on actual chamber geometry	Tube only, steady-state	Full 3D, pulsatile
Cardiac output verified	Not completed	Flow loop measurement
Ejection fraction verified	Not completed	CFD + loop
Cardiac index (BSA-normalised)	Not used	Age-specific targets
Physical experimental validation	None	Flow loop
Right ventricle	Not modelled	Loop can address balance

Thrombosis / stagnation

Not assessed

CFD residence time

The strongest narrative for your project is probably: *"this paper demonstrates feasibility of size but fails to validate the hemodynamic and haematological performance of a downsized TAH — we provide the first experimental and quantitative CFD evidence of those outcomes."* That's a clean and defensible research gap framing.

MIKAELA

SEVAN

INTRO

-> What project is about:

-> What is a total artificial heart

-> Briefly explain heart anatomy (ventricles, atria, etc).

-> Why is it needed (talk about lack of heart donors... needed as BTT option... an example is biventricular heart failure...)

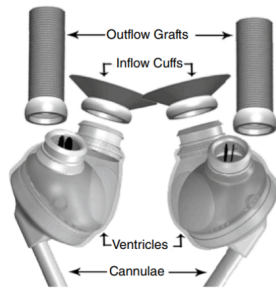
-> An example of TAH is Syncardia (briefly explain FDA approved, pulsatile airflow from external driver, etc)

-> Talk about physical requirements to be a candidate for TAH (BSA + T10)...

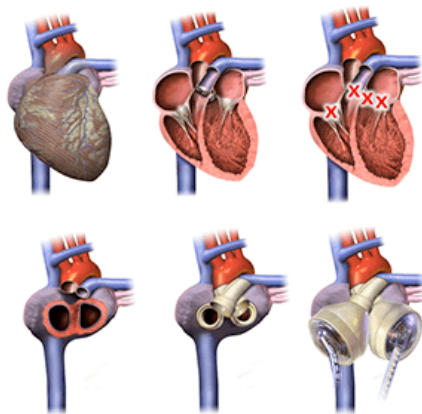
-> Talk about GAPS IN KNOWLEDGE, specifically how people are not eligible for 70cc and 50cc (could talk about parameters for 50cc)... but some people who cannot be eligible includes pediatrics, people with chest disease, etc.

INTRO UPDATED:

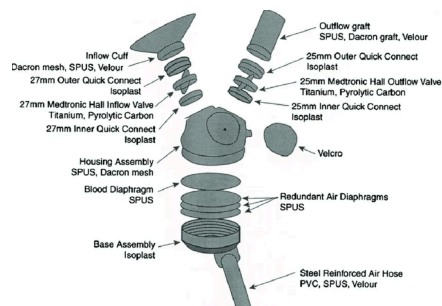
- ~~Firstly, what is heart failure... say how heart needs to pump blood (HH)~~
- ~~Briefly explain heart anatomy... how the heart functions (introduce atria, ventricles, maybe how our diaphragm contraction/expansion causes heart function)~~
- Explain what MCS is... system to assist heart or lung functions.
- Used as bridge to transplant or be temporarily replacement to destination therapy
- ~~TAH are a specific MCS that functions when the majority of the native heart must be replaced, usually used when failure on both sides of the heart.~~
- Last resort... often have acute or irreversible biventricular heart failure with a high risk of death
- We will focus on SynCardia Total Artificial Heart (only FDA approved heart... used in patients with end stage biventricular heart failure)
- ~~Other reasons it might be evaluated for patients... in anticipation for heart transplants, other diagnosis like cardiomyopathy (heart muscle difficult to pump blood), myocarditis (heart muscle inflammation that causes muscle weakness) and other diseases.~~
- “ This is a pulsatile flow total artificial heart that uses an external pulmonary driver.” ... pull up schematic of TAH
- List of potential schematics to use:
- Briefly explain Syncardia, how it consists of 2x ventricle chambers that house blood... with each ventricle having in/outlet valves and a diaphragm base.



- Explain how during implantation... both native ventricles are removed... inflow cuffs are attached to respective atria and outflow grafts attached to the vena respective artery/veins.



- Further describe Syncardia including: shape of ventricle + material (SPUS outer shell, inner dacrom mesh), valve description + material (titanium housing, pyrolytic carbon disc, single pivoting disc that controls blood flow), bottom deforming diaphragm encased in a SPUS base assembly, PVC/SPUS air hose that connects to external driver.



- See visual diagram of Syncardia hooked in patient. Ventricle chambers attached, hole in chest for air hose + external portable driver. External driver acts as our native diaphragm, to move the SPUS diaphragm in ventricle and cause blood flow. Another image of it IRL



- There are also limitations & issues with TAH
- Infection... leading cause of adverse reactions in TAH. Mostly due to driveline exit for the external driver
- Stroke...
- Bleeding...
- Haemolysis...

-> Lead this into students who attempted 35cc heart... however this opens up more research questions... specifically their method of validation was not that good (further explain!!!)

-> This lead to the overall goal of the project: **How downsizing methodology SynCardia Total Artificial Heart affects flow characteristics / haemolysis**

- we apply this to the 35cc study largely

-> This requires CFD studies...

HYPOTHESIS:

- Briefly explain what haemolysis is, measured in NIH -> ASTM equation
- From literature we have observed a trend that heart size decreases, fluid shear increases, (..as well as blood pressure?) for a given flow rate, increasing hemolysis risk

PARAMETERS:

- Flow rate === 5L/min etc

follows:

Steady state flow

Parameter	Role	Why
Volumetric flow rate (5 L/min CO)	Requirement + Input	This is cardiac output at rest on an average, healthy adult. This will be described directly as an input boundary condition.
Mean arterial pressure (180-200mmHg)	Requirement + Output	This is systole in the left ventricle and an output solver will compute. This is a target output that may vary with ventricle geometry changes
NIH (haemolysis index)	Output	This is the dependent variable that is calculated from simulation pathlines and results.

- Mention how our scope is purely for left ventricle, because it operates at a higher pressure compared to right ventricle

METHODOLOGIES (How we are doing this)

- Go over all of the 2D/3D steps (2D to validate the concept... further improving mesh and analysing trends in haemolysis as you change flow rate and relate to hypothesis)

of increasing haemolysis as ... etc NOTE WE CANNOT COMPARE NUMBERS TO 3D LITERATURE, ONLY TRENDS. Then doing 3D steady state.. THIS ONE WE CAN COMPARE NUMBERS TO LITERATURE).

- We will use ANSYS CFX, briefly explain current 50cc CAD shape and how it is informed by literature
 - Explain method of downscaling... say that we will look @ literature more... point out TAH parameterised features such as chamfer etc.(Talk volumetric vs geometric downscaling).
 - Then Sevan to go into pathline stuff **hemolysis (cfd exclusive!!)**
-
- Hey transient sims are like. ...really good !! but only if we can we *actually do it !!!*

INTRO

(introduce ourselves ! supervisor is Rajnish Sharma)

Firstly, what are total artificial hearts? Total artificial hearts are a form of mechanical circulatory support system - being a device that assists the heart in pumping blood through the body.

HELLO THIS IS THE OLD SCRIPT

I'm just gonna go off the new bullet points ive made in the 'conference plan' and time what i currently have. ohhhhhhh

How the heart functions: deoxygenated blood enters the heart from the body through the Vena cava. It enters the right atrium, to the right ventricle, oxygenated in the lungs, re-entering to the left atrium, left ventricle then leaving the body.

Total artificial hearts are used as bridge to therapy device for patients with bi-ventricular heart failure, waiting for a heart transplant as a last case, temporary solution.

Our project focuses on the Syncardia Total Artificial Heart, the only FDA approved TAH. This is a diaphragm driven, pulsatile flow total artificial heart that uses an external pulmonary driver.

I've more or less done the intro (m

Prior to implantation, patients must have the required BSA (Body Surface Area) and T10 measurements. These are metrics to determine if patients are eligible for Total Artificial Heart usage. The way BSA is calculated takes into account both patient height and weight, while T10 checks if the device can physically fit in the patient.

Syncardia have 70cc and 50cc variants. The 50cc variant is eligible for patients with a BSA less than 1.85m^2 and a T10 measurement less than 10cm. A gap in knowledge we've identified are patients not eligible for the 50cc Syncardia - including pediatrics, patients with chest abnormalities, etc.

METHODOLOGY - 2D SIMS

We will be using ANSYS CFX for haemolysis comparison and flow patterns, largely finding high areas of shear and pathline tracking.

We will start with 2D steady state simulations of a simplified 50cc model. This will act as a validation for proof of concept - basically that we can use CFX for haemolysis analysis. We will simulate blood flow at varying flow rates, comparing these results to literature to validate the simulation. We will also use 2D for mesh study and validation - refining mesh to check for element quality, skewness... ensure simulation reliability. We compare trends and not direct NIH numbers because data from 2D simulations is not applicable to 3D literature simulations.

Areas where mesh quality would be most important include around valves and diaphragm boundaries - as these are the areas that experience the highest shear stress. This will also determine the correctness of our hypothesis - whether haemolysis increases with flow rate.

These slides show our current 2D simulation files, the first one shows the diaphragm at full extension during peak fluid outflow, the second shows the diaphragm at full contraction during peak fluid inflow. We chose these two diaphragm positions as they are at the cycle's extremes - simulation greatest potential shear throughout the cycle.

These part files are informed by literature and include simplifications as content is IP locked.

- Simplifications include how valves shall be rectangular blockages, assuming most shear occurs due to the valve lid itself (shear from hinges are negligible at this stage).
- Assume closed valve will be wall boundary conditions - assumption that there are no valve leakage during closed position, or that it is negligible at this stage.
- Parametric features we will change adjust include the graft fillets (areas where the ventricle chamber connect to the outflow tubes). This is important because the abrupt change in geometry can create flow disturbances, large change in velocity and flow separation... all leading large potential shear.
- In addition, there are current conflicts in the part geometry where the valve lid clashes with the diaphragm at full ejection... changing so would affect shear potential.

The specific simulation settings, for example boundary conditions, etc... shall require further research. Because of this, this the initial 2D simulations will be an iterative process.

METHODOLOGY - 3D SIMS

We will move on to 50cc 3D steady state simulations. Being 3D, these simulations are a better reflection of literature - therefore we are able to compare both haemolysis trends and specific simulated NIH values with literature.

When our 50cc simulations are completed, we will created a downscaled 35cc model - comparing these simulation results with the previous 50cc to analyse how downscaling affects shear .

INTRO

1. What are total artificial hearts... MCS... assist pump blood
2. Explain how heart functions: right atria -> ventricle.... left ventricle..
3. BTT for bi-ventricle heart failure... waitlist transplant
4. Focus on Syncardia... FDA approved... diaphragm, pulsatile flow
5. Must have BSA + T10 measurement for elligibility... height/weight... physicall fit
6. 70cc and 50cc variant... list measurements... GAP is identified patients not elligible for 50cc... pediatrics and chest wall deformities

METHODOLOGY - 2D

1. Using ANSYS CFX for haemolysis comparison, flow patterns and large areas of shear.
2. 2D steady state sim... validation of proof of concept.
3. Sim blood flow at varying flow rates... comparing trend we see to trend in litereature.

4. Mesh study and validation (element quality & skewness)... simulation reliability.
5. Not compare NIH results directly bc 2D comparison NOT applicable to 3D literature.
6. Mesh quality importance... valve & moving diaphragm... areas with highest shear stress.
7. 2D sim... test hypothesis of increasing haemolysis with increasing flow rate.

8. Current 2D sim files... explain

9. Chose positions... cycle extremes... greatest potential shear

10. Simplification & assumptions:

- Valve rectangle... most shear on lid (hinge negligible)
- Closed valve is wall (leakage negligible)
- Parametric feature... graft fillet... geometry change... velocity change & flow separation... affect potential shear.
- There is current valve conflict... change hinge position.

Iterative process. Sim settings need further research

METHODOLOGY 3D

1. 50cc 3D steady state simulations
2. 3D better reflect literature... compare trend AND NIH results
3. 50cc complete... downscale to 35cc part. Compare 2 sims... analyse how downscaling method and design affects shear and haemolysis potential.

After KNOWLEDGE GAPS

- Therefore there's a **clinical need** for downsized TAH systems **SLIDE 3**
 - **Study for hypothetical 35cc SynCardia** in 2020 by The University of Akron
 - This study **verifies the anatomical feasibility** for fitment in 6-10 year old patients based on cardiac volume, approximating to a **T10 distance of ~7.5cm** ([src, not verified with paediatric bsa/t10 regression lol](#)) (crucial anterior-posterior dimension for fitment)
 - Analysed flow in the inlet/outlet tubes without valve or diaphragm dynamics, leading us to **ask the question of if this TAH model can provide the required cardiac output while maintaining acceptable hemodynamics and flow characteristics**
- **SLIDE 4** Overall goal of the project is to investigate how **downsizing methodologies in a SynCardia TAH affects flow characteristics and hemolysis, largely in the context of 35cc ventricle volume, to explore if the design is feasible**
- **Hemolysis SLIDE 5**
 - **Damage of red blood cells**, which is a **key issue in mechanical circulation** like with TAHs, releases their oxygen-carrying hemoglobin into the blood plasma, which in high amounts can lead to fatal loss of oxygen delivery to organs. The chronic symptoms of moderate hemolysis for TAH patients are very negative for quality of life.
 - Typically calculated as the **ratio of free hemoglobin to total haemoglobin, the hemolysis index % !!**
 - However in the absence of real blood testing, **we approximate this value from CFD, through measuring the scalar shear stresses of the flow**, because fluid shear is difficult to measure irl
- **We predict** that as we decrease the geometry of the heart, the fluid shear and blood pressure for a given flow rate will rise, increasing hemolysis risk
- **SLIDE 6** Our CFD studies will have the following parameters to test for this
 - Our scope is left ventricle testing only because it's the highest pressure/flow ventricle, its most crucial for hemolysis
 - Flow rate will be tested up to 5L/min, common cardiac output for native hearts
 - Ventricular pressure of 180-200mmHg, which is the systole stroke
 - Output is scalar shear stress, pressures, and flow characteristics across the geometry of our TAH model

METHODOLOGIES

- **Downscaling methods ! SLIDE 9**
 - **Geometric vs volumetric** → V proportional to L^3 , a scale factor of 0.7 the length, results in a 0.34 scaling of volume
 - We have **gentle gradations/chamfers** around the inlets and outlets, those are not parameterized by the literature and we can investigate their effect on fluid characteristics

- **Hemolysis methodologies SLIDE 10**
 - **Heuser–Opitz power-law model** for % of damaged red blood cells, which relates shear stress to exposure time with empirically derived constants
 - But wait ! given the variation in flow, the power law equation breaks down
 - we will need to **sum exposure times** for multiple shear values experienced at different points in the flow
- **SLIDE 11** We will do this by **exporting the pathlines of the flow** and breaking them into fixed time segments with local sheer for that region !! we will then get a more accurate average hemolysis index percent
- We can **compare this dimensionless value to literature** by approximating it to real blood damage with the **Normalized Index of Hemolysis !!**

Notes: ADD SLIDE NUMBERS PER BULLET / TABULATE

- **Heart disease** is becoming **increasingly common**
- End-stage **heart failure renders the native heart unviable** for pumping blood, the **gold standard therapy is a heart transplant**
- **Donor hearts** do not meet this demand, with **long wait times** for receiving patients resulting in death
- **TAHs** are gaining traction as a legitimate biomedical technology, offering an as a **bridge to transplant therapy** (for transplant later) or **destination** (forever) **therapy** for contraindicated patients with antibodies against donor hearts
- One such TAH is the **SynCardia**, the only **FDA-approved TAH**, it works by replacing each ventricle with a chamber that pumps blood with a **pneumatically actuated diaphragm**, connected by an externally-worn driver. It is **pulsatile** (like a beating heart) and has two **valves** per chamber.
- **Limitations** exist for the SynCardia, it's available as medium 50 cubic centimetre (cc) and large 70cc volume, requiring **adequate room/anatomical compatibility** inside your chest (thoracic cavity) to be implanted successfully
 - **Thoracic cavity distance** between your tenth spinal disk and your ribcage (T10 distance) must be greater than **10cm** to fit the (50cc) device, generally ok, but access issue for pediatrics or people with chest deformities
 - **Hemolysis** is a form of **blood damage caused by mechanical stresses**, which are caused in excess by TAHs. Hemolysis increases clotting risk, and requires blood thinners to prevent stroke, but exacerbates existing bleeding, leading to a mortality risk!!

- “So what’s the catch?” I hear you thinking. Why don’t we all just get a SynCardia ? **SLIDE 4 LIMITS**
- The technology still has a lot of limitations, it’s available only 50 cubic centimetre (cc) and 70cc chamber size, requiring **adequate room/anatomical compatibility** inside your chest (thoracic cavity) to be implanted successfully
 - **Thoracic cavity distance** between your tenth spinal disk and your ribcage (T10 distance) must be greater than **10cm** to fit the (50cc) device, generally ok, but access issue for pediatrics or people with chest deformities
 - **Hemolysis is a major issue.** It’s a form of **blood damage caused by mechanical stresses**, which are caused in excess by TAHs. Hemolysis increases clotting risk, and requires **permanent use of blood thinners** to prevent **stroke**, but **exacerbates existing severe TAH side effects** such as **bleeding**, leading to unwanted mortality risk!!
 - These side effects contribute to a **survival rate** that is not fantastic, at around **74% survivorship** in the first month and **51% at 12 months**
- **SLIDE 5 CLINICAL NEED** Therefore there’s a **clinical need** for downsized TAH systems
 - **We predict that device size and hemolysis severity are closely connected**, making this tricky
 - We’ve been inspired by this **study for a hypothetical 35cc SynCardia** in 2020 by The University of Akron
 - This study **verifies the anatomical feasibility** for implantation in 6-10 year old patients, approximating to a **T10 distance of ~7.5cm** ([src, not verified with paediatric bsa/t10 regression lol](#)) (crucial anterior-posterior dimension for fitment), which is a huge jump from the 10cm requirement of the smallest approved SynCardia today
 - **BUT** the study opens **unanswered** questions still for if the device can perform appropriately at this volume, and operates at acceptable hemolysis risk in doing so
- **SLIDE 6 AIM**
 - **Overall goal of the project** then is to investigate how **downsizing methodologies in a SynCardia affects flow characteristics and hemolysis**, largely in the context of a 35cc ventricle volume, exploring the feasibility of this TAH design as an appropriate pumping system

SLIDE 10 DOWNSIZING

- Hi its me again
- Downsizing isn’t as simple as putting a multiplier on the SynCardia CAD file
- There’s a **complex relationship** here that will affect **hemolysis risk**, as **smaller flow volumes increase sheer**, which is the **driving variable for calculating hemolysis**
- We’ll look at **Geometric vs volumetric** scaling → **V is proportional to L³**, making a scale factor of 0.7 the length result in a 0.34 scaling of volume, which could unnecessarily raise hemolytic potential **giving us choices** for what we want to minimize and maximize when scaling

- **Parameterized areas** are also subject to downscaling variations, as discussed earlier
- We have **gentle gradations/chamfers** around the inlets and outlets, those are **not parameterized by the literature** (IP-locked) and we can **investigate their effect on fluid characteristics by varying their dimensions**
- **SLIDE 11 and 12 PATHLINES**
 - To ultimately calculate hemolysis, we will use **CFX post-processing** to **export simulation pathline data** with attached shear, to simulate each pathline **as a blood cell experiencing local shear** that we will sum for the average hemolysis, this data will give us a **clinically comparable value of Hemolysis Index percentage**, and can approximate a normalized index for real damage, **answering our investigation questions**